# Fixed-Point FPGA Implementation and Verification of a Hybrid PSO–Fuzzy MPPT Controller

Fernando Antonio Marques Schettini
*Computer Engineering*
*Centro Universitário SENAI CIMATEC*
Salvador, Brazil
fernandoschettini@outlook.com

Ramon Enrico Alves Papes
*Computer Engineering*
*Centro Universitário SENAI CIMATEC*
Salvador, Brazil
ramonpapes@gmail.com

Davi Santana Cerqueira
*Computer Engineering*
*Centro Universitário SENAI CIMATEC*
Salvador, Brazil
daviscerq04@gmail.com

Caio André Fahel
*Computer Engineering*
*Centro Universitário SENAI CIMATEC*
Salvador, Brazil
caiofahel@gmail.com

Ângelo Ribeiro Meirelles de Oliveira
*Computer Engineering*
*Centro Universitário SENAI CIMATEC*
Salvador, Brazil
angelo.oliveira@aln.senaicimatec.edu.br

Maria Julia Novaes Cerqueira
*Electrical Engineering*
*Centro Universitário SENAI CIMATEC*
Salvador, Brazil
maria.cerqueira@aln.senaicimatec.edu.br

João Alberto Castelo Branco Oliveira
*Centro Universitário SENAI CIMATEC*
Salvador, Brazil
joao.o@doc.senaicimatec.edu.br

***Abstract*—Hybrid Maximum Power Point Tracking (MPPT) algorithms combine global optimization and local control strategies to address the nonlinear behavior of photovoltaic systems under partial shading conditions. However, their implementation on reconfigurable hardware introduces challenges related to numerical representation, resource utilization, timing constraints, and functional verification. This paper presents a modular fixed-point VHDL implementation of a hybrid Particle Swarm Optimization (PSO) and fuzzy logic MPPT controller with Fokker–Planck-inspired adaptive stepping, targeting an Intel Cyclone V FPGA. The architecture integrates measurement processing, adaptive best-solution tracking, fuzzy refinement, and particle updates. Hardware feasibility is investigated through synthesis and Static Timing Analysis (STA), while functional properties are evaluated using a UVM-based testbench comprising 28 scenarios. Reported synthesis results show a utilization of 3,535 Adaptive Logic Modules (19%) and 9 DSP blocks (14%), while timing analysis across four operating corners indicates a worst-case maximum frequency of 5.848 MHz for the controller clock domain, exceeding its configured frequency of 3.125 MHz. Additionally, an open-loop evaluation using 15 months of historical measurements from the Mesa Del Sol microgrid characterizes duty-cycle variability and internal computational behavior; in open loop, these results reflect the fuzzy refinement path, since the PSO state does not feed back into the refined command. The results support the feasibility and initial verification of the proposed FPGA implementation, while closed-loop GMPP tracking performance and photovoltaic energy extraction remain outside the scope of the experimental validation.**

***Index Terms*—MPPT, FPGA, VHDL, fuzzy logic, particle swarm optimization, photovoltaic systems.**

## I. Introduction

Photovoltaic (PV) systems have become a key solution for distributed electricity generation driven by the urgent need for clean energy sources. However, the power extracted from a PV module is directly affected by varying environmental conditions, such as irradiance and temperature. This makes Maximum Power Point Tracking (MPPT) techniques essential to continuously drive the system toward its peak efficiency point.

Under Partial Shading Conditions (PSC), caused by clouds, vegetation, or building shadows, the power-voltage curve develops multiple peaks, resulting in several Local Maximum Power Points (LMPPs) and a single Global Maximum Power Point (GMPP). Under these conditions, classical MPPT methods, such as Perturb and Observe (P&O) and Incremental Conductance, tend to get trapped in the first LMPP encountered, leading to significant power losses and tracking instability. This limitation has driven research toward more sophisticated approaches, particularly those combining population metaheuristics, such as Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO), with intelligent control techniques like fuzzy logic to explore the search space globally while maintaining local stability.

Field-Programmable Gate Arrays (FPGAs) provide a suitable platform for implementing these controllers. FPGAs support parallel execution, deterministic scheduling, and field reconfigurability for post-deployment updates without dedicated silicon fabrication. These features motivate their use in PV applications, including remote off-grid locations where hardware adaptability is useful.

Recent literature highlights the potential of these approaches while revealing gaps within the reviewed corpora. Lima and Oliveira reviewed fuzzy logic controllers in FPGA for MPPT, identifying opportunities for their combination with metaheuristics [1]. Similarly, the review by Cerqueira and Oliveira

examined 37 studies in detail from 89 selected IEEE Xplore articles on MPPT under partial shading [2]. It reported efficiencies between 99% and 99.9% with tracking times of 0.05 to 0.4 s for P&O and GWO combined with metaheuristics, compared with 85–90% efficiency and 0.5–2.0 s for classical approaches; P&O combined with intelligent control showed wider ranges of reported values. These ranges reflect heterogeneous test conditions and do not establish a controlled ranking. Only 2 of the 89 selected studies explicitly reported FPGA implementation; standardized hardware verification and static timing analysis were not identified in the 37-study detailed subset. These observations apply to the examined corpus, whose selection limitations are discussed in Section III.

In this context, Patnaik et al. proposed a hybrid algorithm combining advanced PSO, fuzzy logic, and a Fokker–Planck-based approach [3]. The present work develops a synthesizable fixed-point FPGA architecture motivated by that approach, documenting its adaptations, specified functional checks, and hardware characterization without claiming priority or exact equivalence to the reference algorithm.

Within the examined literature, three implementation needs motivate this work:

(i) the scarcity of FPGA-based implementations of hybrid PSO-Fuzzy MPPT controllers using full fixed-point arithmetic;
(ii) the lack of functional verification based on standardized hardware methodologies, such as the Universal Verification Methodology (UVM), for this class of controller; and
(iii) the absence of static timing analysis (STA) characterization to complement synthesis validation.

These gaps motivate the central research question of this work: *Can a hybrid PSO–Fuzzy MPPT controller with Fokker–Planck-inspired adaptive stepping be implemented as a synthesizable fixed-point FPGA architecture while satisfying defined functional and timing requirements?*

To address the research question, the main objective of this work is to implement, verify specified properties of, and characterize a hybrid PSO-Fuzzy MPPT controller architecture with Fokker–Planck-inspired adaptive stepping, fully described in synthesizable VHDL with fixed-point arithmetic and motivated by Patnaik et al. [3]. The specific objectives are defined as follows:

- Implement a modular, synthesizable VHDL architecture using fixed-point arithmetic, incorporating adaptations for numerical processing and open-loop operation;
- Evaluate FPGA resource utilization and perform Static Timing Analysis (STA) on an Intel Cyclone V device;
- Verify functional properties using a UVM-based testbench covering normal operation, numerical boundaries, and adverse inputs;
- Characterize open-loop computational behavior using 15 months of historical data from the Mesa Del Sol microgrid, analyzing duty cycle variability and internal controller signals.

The main contributions of this work are:

- A modular fixed-point VHDL implementation of the hybrid PSO-Fuzzy controller with Fokker–Planck-inspired stepping, incorporating documented adaptive-memory and numerical adaptations, with synthesis reporting 9 of 66 DSP blocks on a Cyclone V FPGA;
- A UVM-based functional verification environment reporting five specified criteria across 28 test scenarios, with V5 limited to the documented reference-power calculation pending confirmation of a DUT comparison;
- Static timing characterization across four operating corners, including a worst-case maximum frequency of 5.848 MHz for the controller clock domain and the controller update latency;
- An open-loop evaluation using historical photovoltaic measurements to characterize the duty-cycle variability of the fuzzy refinement path and the computational behavior without claiming closed-loop MPPT performance.

The historical-data assessment was conducted in open loop: generated commands did not affect subsequent voltage or current inputs. It complements hardware characterization and precedes closed-loop evaluation with a PV array and DC–DC converter.

The remainder of this paper is organized as follows: Section II presents the theoretical background. Section III describes the systematic literature review. Section IV details the hybrid controller architecture and hardware implementation. Section V presents the UVM-based functional verification. Section VI addresses static timing analysis. Section VII discusses the experimental results and descriptive metrics. Finally, Section VIII concludes the paper and outlines future work.

## II. Theoretical Background

### A. Nonlinear Photovoltaic Modeling and Partial Shading

Photovoltaic (PV) systems exhibit nonlinear current–voltage (I-V) and power–voltage (P-V) characteristics, which depend directly on environmental conditions such as solar irradiance and cell temperature. To maximize conversion efficiency, the system must operate at its Maximum Power Point (MPP). However, under Partial Shading Conditions (PSC), the P-V curve develops multiple peaks, resulting in several Local Maximum Power Points (LMPP) and a single Global Maximum Power Point (GMPP) [4].

Fig. 1 illustrates this nonlinear behavior, highlighting the multiple operating points on a P-V curve under partial shading. As can be observed in the figure, the main control challenge lies in bypassing the smaller peaks (LMPP) to reach the highest peak (GMPP). MPPT techniques that fail under these conditions trap the operating point at the first LMPP found, which results in energy generation losses and justifies the use of global heuristics.

### B. Fuzzy Logic

Fuzzy logic is an intelligent control technique that does not require a rigorous mathematical model of the system, making it suitable for use in nonlinear systems. In MPPT algorithms,

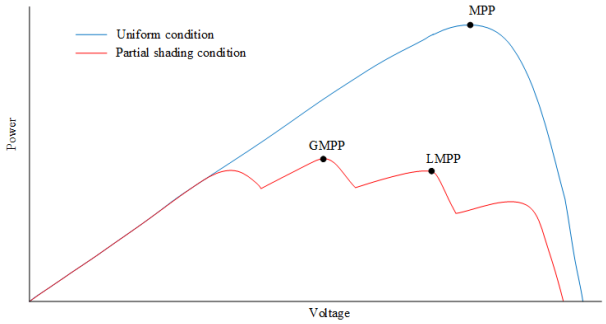



Figure 1. P-V curve under partial shading conditions showing the presence of LMPPs and a single GMPP [5].

the main objective of the controller is to drive the system to the point where the variation of power with respect to voltage is zero, that is, $dP/dV = 0$.

To map this dynamic behavior and feed the inference system, the concepts of error $E(k)$ and change of error $\Delta E(k)$ at sampling instant $k$ are defined. The error is expressed by the discrete approximation of the derivative of power with respect to voltage, as shown in Eq. 1:

$$E(k) = \frac{P(k) - P(k-1)}{V(k) - V(k-1)} \tag{1}$$

The change of error, in turn, describes the change in this finite-difference gradient between two consecutive samples. Neither this signal nor a zero gradient alone identifies the GMPP under partial shading. This signal is formalized in Eq. 2:

$$\Delta E(k) = E(k) - E(k-1) \tag{2}$$

where $P(k)$ and $V(k)$ represent the instantaneous power and voltage generated by the panel in the current period, while $P(k-1)$ and $V(k-1)$ correspond to the respective values recorded in the immediately preceding sampling cycle.

Fig. 2 details the internal architecture and membership functions of a standard fuzzy controller based on these two variables. The pipeline of this controller typically operates in three stages: fuzzification, responsible for converting the numerical variables obtained from Eqs. 1 and 2 into degrees of membership in linguistic sets; inference, which combines these degrees in a base of IF-THEN logical rules to determine the corrective action; and defuzzification, which converts the linguistic outputs back into a continuous physical signal to adjust the converter *duty cycle*.

Although it offers robustness and fast dynamic response for uniform irradiance variations, the performance of the fuzzy controller is intrinsically dependent on the heuristic formulation of its rules and on the tuning of the membership functions, and it may show limitations in the global search under partial shading [1].

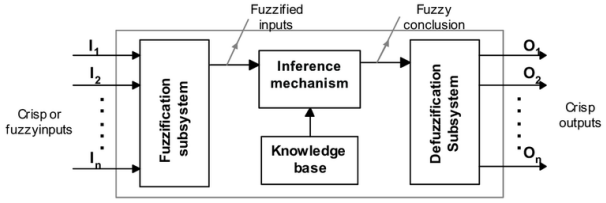



Figure 2. Typical structure of a fuzzy expert system containing the processing blocks and the variable mapping loop [6].

### C. Particle Swarm Optimization

Particle Swarm Optimization (PSO) is a stochastic metaheuristic algorithm based on swarm intelligence, inspired by the social behavior of bird flocks and fish schools, originally introduced by Kennedy and Eberhart [7]. In the context of maximum power tracking, each particle represents a candidate solution, such as the value of the converter *duty cycle*. The particles move through the search space by updating their velocities and positions based on their best individual experience ($p_{best}$) and on the best experience of the whole swarm ($g_{best}$).

The velocity update $v_i^{k+1}$ of particle $i$ at iteration $k$ is calculated as shown in Eq. 3, while the update of its respective position $x_i^{k+1}$ occurs as shown in Eq. 4:

$$v_i^{k+1} = w v_i^k + c_1 r_1 (p_{best,i} - x_i^k) + c_2 r_2 (g_{best} - x_i^k) \tag{3}$$

$$x_i^{k+1} = x_i^k + v_i^{k+1} \tag{4}$$

where $w$ is the inertia weight, $c_1$ and $c_2$ are the cognitive and social acceleration coefficients, respectively, and $r_1, r_2$ are random variables uniformly distributed in the interval $[0, 1]$. The implementation described in Section IV departs from this formulation by replacing $r_1, r_2$ with pseudorandom factors $\rho_1, \rho_2$ restricted to a narrow configured interval.

## III. Systematic Review

To identify technological trends in addressing partial shading conditions (PSC) in photovoltaic systems, with emphasis on FPGA-based approaches, a systematic literature review was conducted on the IEEE Xplore database. This section summarizes the adopted protocol, the results obtained, and the gaps identified.

### A. Review protocol

After evaluating approximately twenty search scopes, the following search *string* was applied to the IEEE Xplore database, returning 117 publications:

```
("partial shading condition" OR PSC)
        AND ("solar panel")
AND (FPGA OR "field-programmable gate array")
```

The selection was guided by one primary research question (PRQ) and two secondary research questions (SRQ):

- **PRQ:** Which techniques, methods, or tools have been applied to mitigate the effects of shading in photovoltaic systems?

- **SRQ1:** Which algorithms have been primarily employed to mitigate the effects of shading?
- **SRQ2:** Which techniques are used to evaluate the performance of the proposed algorithms?

The inclusion (IC) and exclusion (EC) criteria applied are presented in Table I. Of the 117 publications, 89 were selected and 28 excluded; among the selected ones, only two explicitly addressed FPGA implementation. The techniques employed in the 89 papers were mapped by reading titles and abstracts, recorded in a spreadsheet, and grouped according to the technique presented as the result. The available review documentation does not specify how the 37 studies examined in greater detail were selected from these 89 publications, limiting the reproducibility and representativeness of that subset.

Table I
INCLUSION AND EXCLUSION CRITERIA OF THE SYSTEMATIC REVIEW.

| Inclusion criteria (IC) | Exclusion criteria (EC) |
|---|---|
| **IC1:** the study presents original results and relevant contributions to the area. | **EC1:** the study does not present satisfactory or relevant techniques for shading mitigation. |
| **IC2:** shading in photovoltaic systems is addressed as a central theme, not merely as a secondary mention. | **EC2:** the paper lacks an adequate contextual review of the investigated problem. |
| **IC3:** the paper is available in full text and presents sufficient technical content to enable in-depth analysis. | **EC3:** the material corresponds to duplicates or preliminary versions of studies already included. |

### B. Results

In response to the PRQ and SRQ1, three main techniques were identified in the 37-study detailed subset: PSO, with 15 papers; P&O, with 14; and GWO, with 8 (Fig. 3). Regarding SRQ2, the studies were summarized using their reported energy efficiency, tracking time (dynamic response), and operational stability, expressed by oscillation around the maximum power point. Table II preserves these reported indicators by family and subgroup; heterogeneous experimental conditions and metric definitions preclude a controlled cross-study comparison.

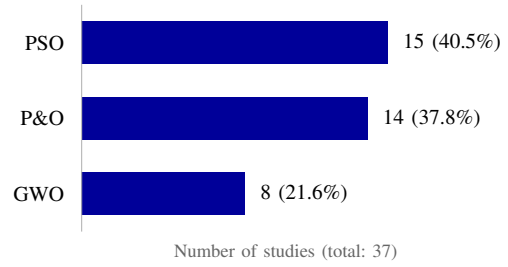



Figure 3. Distribution of the 37 studies among the three main techniques identified in the systematic review.

*PSO:* of the 15 studies, eleven were conducted exclusively through simulation, two employed real prototypes, and two

Table II
REPORTED PERFORMANCE BY ALGORITHM FAMILY UNDER PARTIAL SHADING.

| Family / group | Eff. (%) | Track. (s) | Oscillation |
|---|---|---|---|
| *PSO* | | | |
| Traditional / improved† | – | 0.08 / 0.0095 | – |
| *P&O* | | | |
| Traditional | 85–90 | 0.5–2 | Pronounced |
| Optimized | 99–100 | 0.04–1.8 | Reduced |
| With intelligent control | 98–113* | 0.01–1.7 | Smooth |
| With metaheuristics | 99–99.8 | 0.1–0.4 | Near zero |
| *GWO* | | | |
| Pure and improved | 99.4–99.8 | 0.09–0.58 | High stability |
| Hybrid GWO–P&O | 99–99.4 | 0.14–0.3 | Minimal |
| With other techniques | 99.8–99.9 | 0.05–0.15 | Excellent stability |

†MATLAB/Simulink simulations; efficiency and oscillation not reported in aggregate form. *Values as reported in the primary studies; values above 100% require clarification of their denominators and are not interpreted as absolute tracking efficiencies.

combined simulation with experimental implementation, so most of the evidence is grounded in simulated scenarios. The best response times were obtained in MATLAB/Simulink: approximately 0.08 s to reach the GMPP in the traditional version and down to 0.0095 s in the improved version [8]. Among the comparative studies, combining PSO with deterministic techniques in *quasi-Z-source* inverters improved DC-link voltage stability and reduced steady-state oscillations [9]; using PSO to locate the GMPP, followed by P&O for local fine-tuning, reduced convergence time and relative error [10]; and the PSO–P&O and PSO–INC hybrids maintained higher efficiency than conventional algorithms under dynamic shading, in both simulation and experimental implementation. In another comparative analysis, PSO reached the GMPP in 0.08 s, whereas ABC achieved a shorter initial time of 0.05 s but required about 0.4 s to stabilize due to oscillations [11].

*P&O:* the 14 studies were grouped into four categories. Traditional P&O exhibited pronounced oscillations around the MPP and a tendency to become trapped in LMPPs [12]. The optimized versions reported reduced oscillations and increased accuracy under their test conditions [13], [14]. The group integrating P&O with intelligent techniques, such as fuzzy logic, the Firefly Algorithm, the Flower Pollination Algorithm, and Fokker–Planck with PSO, reported smooth oscillations, high stability, and fast convergence under irradiance variations [3], [12], [15]. Finally, combinations with evolutionary metaheuristics, such as PSO, GWO, MGWO, MIWO, and ABC–WOA, reported favorable indicators and practically zero *ripple* in the respective studies [16], [17].

*GWO:* the eight studies were classified into pure and improved GWO [18], [19], hybrid GWO–P&O [20], and GWO combined with other optimization techniques, such as BGWO, GWO–PSO, and GWO–GA [21]. The latter group reported high efficiencies and short tracking times under its respective test conditions, without establishing superiority across all

evaluated groups.

### C. Discussion

The PSO family reported short tracking times in optimized versions and favorable results for hybrid implementations; however, most studies are restricted to simulations, and performance is sensitive to the tuning of the inertia and acceleration coefficients, making adjustment dependent on the operating scenario. For P&O, the traditional version showed limited performance under shading, whereas optimized, intelligent, and metaheuristic-hybrid variants reported improvements in their respective studies. This evidence relies heavily on simulation, highlighting the need for standardized metrics, practical validation, and normalization of efficiency criteria. For GWO, pure and improved versions and their hybrids reported favorable tracking behavior, but comparisons with PSO depend on the tested configuration. Hybrid GWO approaches reported low *ripple*, high GMPP tracking accuracy, and adaptation to irradiance variations in the reviewed studies, at the cost of greater computational complexity and more parameters to tune, with limited validation on real prototypes. These findings motivate hardware investigation without establishing a universal ranking of algorithms.

### D. Limitations and research opportunities

Within the examined corpus, the review identified three main limitations: (i) the predominance of studies based on simulation or bench-top prototypes; (ii) heterogeneous test conditions, including PV array configurations and converter architectures, which hinder direct comparisons and the transfer of results to field operation; and (iii) limited documented use of FPGA hardware, with only two of the 89 selected studies explicitly implementing their solutions on this platform. This count does not estimate the total number of FPGA implementations in the scientific literature.

The review itself is limited to IEEE Xplore and the stated search expression. Terms such as "solar panel" and the required FPGA-related terms may omit relevant studies using alternative terminology. The relevance and adequacy judgments in Table I introduce subjectivity, while the undocumented reduction to 37 detailed studies further limits reproducibility. These constraints and the heterogeneous evaluation conditions restrict the generality of the conclusions.

Based on these limitations, the review points to the following opportunities: experimental field validation, with modules exposed to real weather conditions and shading patterns; the proposal of a *benchmark* with uniform scenarios, parameters, and indicators, accompanied by a public dataset containing real and simulated shading profiles; optimized implementations of classical and advanced algorithms on FPGA, evaluating latency, power consumption, and architectural flexibility, as well as adaptive controllers with dynamic reconfiguration; hybrid and intelligent algorithms with low computational cost, suitable for FPGA execution and equipped with explainability mechanisms; and complete systems that incorporate, in addition to MPPT, monitoring, diagnostics, and fault detection, considering module degradation, soiling, mismatch, and long-term reliability.

### E. Relation to the present work

The results of the systematic review motivate this work in two respects. First, the small number of explicitly reported FPGA implementations in the selected corpus (2 out of 89 studies) motivates the synthesizable VHDL description, resource characterization, and static timing analysis presented in Sections IV and VI. Second, the reported behavior of hybrid approaches, including PSO with a fuzzy Fokker–Planck solution [3], motivates the choice of reference approach without establishing its superiority under common conditions. Experimental validation under real conditions remains a future step, as discussed in Section VIII.

## IV. Architecture of the Hybrid PSO–Fuzzy MPPT Controller

Fig. 4 presents the algorithmic flow of the hybrid PSO-Fuzzy controller motivated by [3]. After initializing the swarm, candidate, and adaptive memory, the controller selects a particle command, waits, and reads voltage and current. It computes scaled power, variations, error, and change of error. Fuzzy inference, adaptive stepping, and P&O refine the previous candidate, while adaptive memory updates *pbest* and recomputes *gbest*.

Sampling and memory updates repeat until all particles have been processed. The controller then defines the search window, generates coefficients, and updates the swarm. It inserts the refined candidate as particle zero, resets that particle's velocity, and starts the next sweep.

The hybrid structure combines fuzzy refinement with PSO candidate updates, with the intended closed-loop role of searching for a maximum power point while limiting command oscillations (*chattering*). In the historical-data evaluation, each iteration updates a duty-cycle command, but this command is not applied to a converter or plant model. Subsequent input measurements therefore remain independent of the command, and particles receive no physical power feedback associated with their candidate duty cycles.

### A. Hardware implementation

The hardware implementation was described in synthesizable VHDL and organized hierarchically and modularly, separating the *datapath* from the control logic. The design consists of a package of common definitions and functions (`hybrid_mppt_pkg`), the main controller containing the state machine (`hybrid_pso_fuzzy_mppt`), six combinational units for measurement, fuzzy refinement, adaptive best-state updates, search bounds, random coefficients, and particle updates, and two top levels: the synthesis top level `mppt_fpga_top`, which adds a clock and reset unit (`mppt_clock_reset_unit`), and the purely structural simulation wrapper `mppt_top`. This separation favors the inference of dedicated arithmetic operators and keeps the controller's sequential logic concentrated in a single synchronous process.

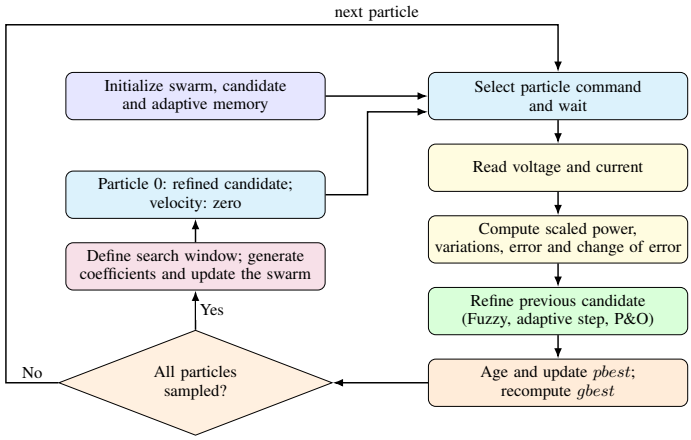



Figure 4. Algorithmic flow of the hybrid PSO-Fuzzy controller. State-update loops do not represent feedback through a PV system in the historical-data evaluation.

Fig. 5 separates the core interface from its internal architecture. Fig. 5a shows the controller instance as synthesized within `mppt_fpga_top`: 16-bit signed voltage and current inputs, clock, reset, and enable, and the two outputs that leave the device, the 8-bit `duty_out`, which carries the current particle command, and `store_valid`. The simulation wrapper `mppt_top` additionally exposes integer observation ports; among them, `control_duty_out` carries the refined candidate used in the reported duty-cycle metrics. These observation ports exist only in simulation and are not outputs of the synthesized design.

Fig. 5b shows seven blocks and their interconnections:

- **FSM, registers, and selection:** stores controller state, selects operands, accumulates *gbest*, and registers outputs.
- `mppt_measurement_unit`: converts voltage, current, and sample history into scaled power, variations, error, and change of error.
- `mppt_fuzzy_ffp_unit`: uses measurement signals and the previous candidate to produce the correction, adaptive step, and refined duty.
- `pso_best_tracker_unit`: decays stored power and returns the selected particle's updated *pbest* position, power, and age.
- `pso_search_window_unit`: maps the search center and configured radius to duty bounds for the particle update unit.
- `pso_random_coeff_unit`: produces one coefficient pair and the next LFSR state from the registered LFSR state.
- `pso_particle_update_unit`: computes velocity and position from selected particle states, *pbest*, *gbest*, coefficients, and search bounds.

The reported synthesis used Intel Quartus Prime Lite Edition 20.1.1, targeting the Cyclone V 5CEBA4F23C7 device. Table III records the resulting resource mapping, including 9 DSP blocks out of 66 (14%), with no block memory or PLL usage. The synthesized top level, `mppt_fpga_top`, instantiates the controller with the hyperparameters selected in Section VII-B and `SETTLE_CYCLES` $= 1000$; since generics are elaborated as constants, the resource figures and the timing results in Section VI refer to this configuration.

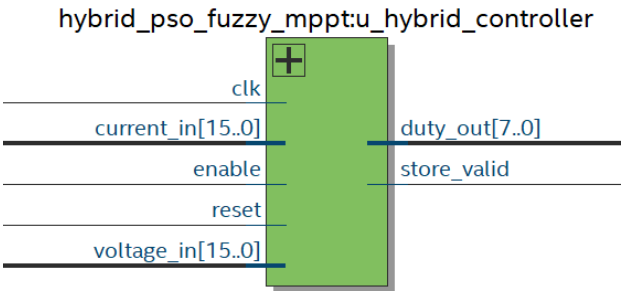
Table III
SYNTHESIS SUMMARY OF THE HYBRID PSO-FUZZY MPPT CONTROLLER.

| Resource / information | Value |
|---|---|
| Synthesis tool | Quartus Prime Lite 20.1.1 |
| FPGA family | Cyclone V |
| Target device | 5CEBA4F23C7 |
| Synthesis status | Successful |
| Logic utilization (ALMs) | 3,535 / 18,480 (19%) |
| Dedicated logic registers | 677 |
| I/O pins | 44 / 224 (20%) |
| Virtual pins | 0 |
| Block memory bits | 0 / 3,153,920 (0%) |
| RAM blocks (M10K) | 0 / 308 (0%) |
| DSP blocks | 9 / 66 (14%) |
| PLLs | 0 / 4 (0%) |
| DLLs | 0 / 4 (0%) |

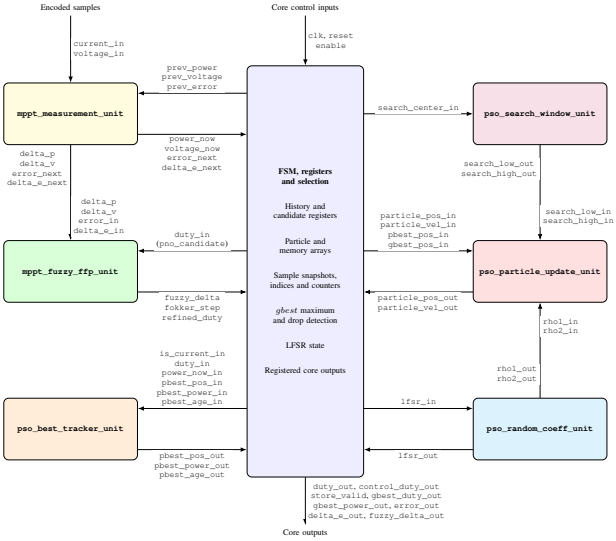
*1) Numerical representation and adaptations with respect to the reference algorithm:* The implementation is motivated by, rather than an established exact reproduction of, the reference algorithm [3]. Numerical adaptations use integers and fixed scaling instead of floating point. Voltage is preprocessed in Q12.4 format, that is, $V_{int} = 16\,V$, and current is normalized in Q1.15 relative to a current base. In the historical-data evaluation, these inputs are not PV-side DC quantities: the voltage is the AC voltage of the microgrid low-voltage switchboard and the current is derived from the PV active power, as described in Section VII. The available preprocessing code rounds and clips encoded inputs to the signed 16-bit range, while VHDL integer divisions truncate toward zero and explicit clamps limit internal signals. The internal power is calculated according to Eq. 5; the divisor 65536 defines its internal scale and does not, by itself, restore power in watts [22].

$$P_{int} = \mathrm{trunc}\left(\frac{V_{int}\, I_{int}}{65536}\right) \tag{5}$$

In closed-loop MPPT operation, the controller acts on a




(a) Controller instance in the synthesized top level `mppt_fpga_top` (Quartus RTL viewer).




(b) RTL architecture with module inputs and outputs labeled on the arrows.

Figure 5. Core interface and module interconnections of the hybrid PSO-Fuzzy MPPT controller. The architecture in (b) is contained in `hybrid_pso_fuzzy_mppt`; arrows identify the data or control signals carried by each connection.

photovoltaic system connected to a DC–DC converter, so that duty-cycle changes affect the panel operating point and extracted power. Here, evaluation used previously measured and preprocessed historical data. The calculated duty cycle is not applied to a physical system or closed-loop electrical model; the input-power sequence is determined by the dataset rather than the generated command. The reported power values refer to the exported scaled input-power signal. Interpreting them in watts or comparing their magnitudes across differently normalized files requires the preprocessing settings and current bases associated with the reported runs.

Consequently, the historical-data evaluation is treated as an open-loop computational assessment rather than a validation of MPPT tracking performance. The experiments characterize the numerical behavior of the implemented architecture, the variability of its generated duty-cycle commands, and its response to changes in the input measurements. In particular, the evolution of *pbest* and *gbest* is interpreted as internal algorithmic state behavior, not as evidence of convergence toward the physical global maximum power point (GMPP).

Since the dataset contains measurements from several days, the sample sequence was processed continuously, without automatic reset of the controller when the date changes. To prevent *pbest* and *gbest* from representing fixed and indefinitely valid historical maxima, an adaptive memory with exponential forgetting was implemented. Thus, the power value associated with the *pbest* of each particle decays before being compared with new samples, and the *gbest* is recalculated from the adaptive *pbest* values. In the final configuration adopted, a *memory half-life* of 450 samples was used, corresponding to a forgetting factor of approximately 0.9985, and a maximum *pbest* age of 500 samples. Negative powers in the dataset are still used in the error calculation, but are not considered as the best power in PSO.

Adaptive memory, exponential forgetting, and changes to the best-state update behavior are algorithmic adaptations. Modularization, FSM scheduling, arithmetic replication or sharing, parameterization, and resource mapping are implementation decisions. These are distinct from the numerical adaptations above and are not asserted to be original algorithmic innovations.

*2) Top levels and hierarchical organization:* The synthesis top level, `mppt_fpga_top`, instantiates `mppt_clock_reset_unit`, which divides the 50 MHz board clock by `CLK_DIV_G` = 16 and synchronizes the reset release to the divided clock, and the `hybrid_pso_fuzzy_mppt` controller. It exposes only the 44 pins needed by the device interface, with `duty_out` and `store_valid` as outputs. The module `mppt_top` is a purely structural simulation wrapper: it exposes the configurable parameters (*generics*), the controller ports, and the observation ports, and instantiates `hybrid_pso_fuzzy_mppt`, passing all signals and parameters through; it is not part of the synthesized hierarchy. Both top levels instantiate the same controller, so the simulated and synthesized logic differ only in the clock and reset unit and in the exposed ports. The `current_in` and `voltage_in` inputs are 16-bit signed vectors (`signed(15 downto 0)`), while the `enable` signal allows the controller's evolution to be suspended without requiring a *reset*. The set of configurable parameters includes, among others, the number of settling cycles, the PSO coefficients, the velocity limits, the search window radius, the Fokker–Planck steps, and the adaptive memory parameters, allowing the controller's behavior to be adjusted without modifying the structural description [3].

*3) Main controller and state machine:* The hybrid_pso_fuzzy_mppt module holds the state register and sequential control logic. Its FSM has seven states for N_PARTICLES $= 10$: APPLY_PARTICLE places the current particle position on duty_out; WAIT_SETTLE inserts a configurable wait with architectural default SETTLE_CYCLES $= 1000$; and SAMPLE_AND_UPDATE registers the error, change of error, fuzzy correction, refined candidate, and measurements. TRACK_BEST processes one particle per enabled clock cycle, accumulating the updated *pbest* maximum; its final cycle registers *gbest* and emits store_valid. After all particles have been sampled, PREPARE_SWARM selects the search center, UPDATE_SWARM updates one particle per enabled cycle using a newly generated coefficient pair, and FINALIZE_SWARM inserts the refined candidate as particle zero with zero velocity. A complete control iteration over the ten particles therefore takes $10\,\text{SETTLE_CYCLES} + 142$ clock cycles, as confirmed in simulation (Section VI). The historical-data testbench uses SETTLE_CYCLES $= 1$ to reduce simulation cost; neither setting establishes physical converter settling. Initial positions use a linear feedback shift register (LFSR) with seed 0xACE1, while initial velocities are zero [3], [7], [23].

*4) Measurement unit:* The combinational unit mppt_measurement_unit is responsible for transforming the voltage and current readings into the quantities used by the control. It calculates the instantaneous power according to Eq. 5, as well as the power ($\Delta P$) and voltage ($\Delta V$) variations with respect to the previous sample, and the normalized error, defined as the ratio $\Delta P/\Delta V$ weighted by a configurable gain. To reduce artificial spikes in the error, very small voltage variations are handled explicitly: when $|\Delta V|$ is below a threshold (DELTA_V_MIN $= 16$), the derivative is considered undefined and the error is forced to zero instead of fabricating an artificial slope, a common situation in quantized historical data. Both the error and the change of error are saturated to the interval $[-100, 100]$, ensuring that the indices of the fuzzy membership functions remain within the expected range [24], [25].

*5) Fuzzy block with Fokker–Planck step and P&O:* The mppt_fuzzy_ffp_unit module, supported by the functions defined in hybrid_mppt_pkg, implements a fuzzy controller with Mamdani-style min inference and singleton rule consequents, defuzzified by the height method, which is equivalent to a zero-order Sugeno scheme. The error and the change of error are each mapped into seven fuzzy sets, described by triangular membership functions and by shoulder trapezoids at the extremes, all evaluated in scaled integer arithmetic from 0 to 100. Inference uses the minimum operation (fuzzy AND) over a $7 \times 7$ rule table (FUZZY_RULES), and defuzzification is performed by the weighted average of the rule outputs by their respective firing degrees, producing the fuzzy correction fuzzy_delta. A configurable dead zone (DEADZONE = 3) sets fuzzy_delta to zero when the absolute error and change of error are both below the threshold. It does not cancel the duty adjustment: the step described below then reduces to its minimum, and the candidate still moves by FOKKER_STEP_MIN in the P&O direction whenever $\Delta P$ and $\Delta V$ are both nonzero, remaining unchanged only when that direction is undetermined. The dead zone thus limits the adjustment magnitude to attenuate *chattering* [1], [26].

An adaptive step inspired by the Fokker–Planck equation is applied to this correction: the magnitude of the fuzzy output linearly modulates the step size between a configurable minimum and maximum (FOKKER_STEP_MIN $= 1$ and FOKKER_STEP_MAX $= 12$), with larger fuzzy-output magnitudes producing larger steps. The adjustment direction is defined by Perturb and Observe (P&O) logic from the signs of $\Delta P$ and $\Delta V$; when this direction is undetermined, the sign of the fuzzy output is used. The refined duty cycle is obtained by adding the product of the direction and the step to the current candidate, with a configurable sense (DUTY_DIRECTION $= -1$), and limiting the result to $[0, 100]\%$ [3].

*6) Adaptive best-solution tracker:* The combinational module pso_best_tracker_unit updates the adaptive memory of one selected particle. At each sample, the power associated with the *pbest* of each particle is multiplied by a forgetting factor derived from the configured *memory half-life*, approximated as $1 - \ln 2/\text{MEMORY_HALF_LIFE}$ and quantized in steps of $10^{-4}$: for 450 samples, the resulting factor is $0.9985$. Because the decayed value is truncated to an integer, the decay is approximately exponential only for larger stored powers; below about 667 scaled units, each sample removes exactly one unit, so low-power memory decays linearly, as visible at the end of the day in Fig. 10b. The best individual power of the current particle is updated when the new power exceeds the already decayed value, or when the age of the *pbest* reaches the maximum allowed age (MAX_PBEST_AGE $= 500$ samples), preventing old optima from remaining valid indefinitely. The FSM applies this unit to every particle per sample and accumulates the updated *pbest* maximum to obtain *gbest*. Negative powers are disregarded for the purpose of the best solution, although they still contribute to the error calculation. The FSM also provides an optional mechanism for detecting abrupt changes in conditions (ENABLE_CHANGE_DETECTION): when enabled, if the power of the current sample remains below DROP_THRESHOLD_PERCENT of the swarm-wide best power for DROP_PATIENCE consecutive samples, all *pbest* records and the *gbest* are reinitialized from the current sample. It is disabled in the selected configuration, so its logic is removed by constant propagation in the synthesized design [3], [7], [23].

*7) Search window and random coefficients:* The pso_search_window_unit unit dynamically defines the exploration region of the swarm from a center and a configurable radius (SEARCH_RADIUS $= 6$), producing lower and upper bounds already saturated to the valid duty cycle interval. The window center is selected by the FSM according to a configurable mode and may follow the refined candidate, the *gbest*, or the average of

both, which allows switching between local and global exploration; the selected configuration centers the window on the *gbest* (`SEARCH_CENTER_MODE` = 1). During `UPDATE_SWARM`, `pso_random_coeff_unit` generates one pair of stochastic PSO coefficients per enabled cycle: a 16-bit LFSR, with feedback from positions 15, 13, 12, and 10, produces, for each particle, two random factors $\rho_1$ and $\rho_2$ restricted to the configured interval (`RHO_MIN` = 55 to `RHO_MAX` = 65, that is, 0.55 to 0.65 after scaling). Each factor is obtained from the low byte of the LFSR state modulo the interval width, so it takes 11 discrete values with a slightly nonuniform distribution; unlike the factors $r_1, r_2 \sim U[0,1]$ of Eq. 3, they vary only within this narrow interval. The coefficients feed the particle update unit directly, and the FSM registers the returned LFSR state after two steps per particle. The use of an LFSR provides a deterministic, low-cost pseudorandom source in hardware, ensuring the reproducibility of the simulations [27].

*8) Particle swarm update:* The controller instantiates one `pso_particle_update_unit` and reuses it across the ten particles during `UPDATE_SWARM`. The unit calculates the new velocity from inertia, cognitive attraction to *pbest*, and social attraction to *gbest*, weighted by `W_PSO`, `C1_PSO`, `C2_PSO` (0.30, 0.30, and 0.50 after scaling in the selected configuration), and the random factors $\rho_1$ and $\rho_2$. Integer products are rescaled by fixed divisions, and velocity is saturated to `[VEL_MIN, VEL_MAX]` $= [-30, 10]$. The new position adds velocity to the current position and is limited to the search window and duty-cycle interval $[0, 100]$. The FSM registers one particle's new position and velocity per enabled cycle; a complete control iteration also includes sampling, best-state tracking, and finalization [3], [7], [23].

## V. Functional Verification with UVM

To complement the synthesizability validation presented in Table III, a functional verification testbench based on the *Universal Verification Methodology* (UVM) [28] was developed. While synthesis in Quartus establishes resource mapping, the reported functional simulations assess specified properties of the VHDL DUT under the applied stimuli, including boundary values, negative inputs, and abrupt transitions. Passing these checks does not establish complete algorithmic correctness, equivalence to an independent numerical model, or exhaustive range and protocol coverage.

UVM is a standardized SystemVerilog library maintained by Accellera that defines a reusable and modular testbench architecture. It is widely adopted in the semiconductor industry because it separates stimulus generation, signal observation, and result checking into independent components, which facilitates the maintenance and extensibility of the test environment [28].

### A. Interface adaptation: VHDL–SystemVerilog wrapper

The original `mppt_top` module uses VHDL `integer` ports for the observation outputs (`control_duty_out`, `gbest_duty_out`, `error_out`, among others). Since SystemVerilog does not support direct connection to the VHDL `integer` type, an intermediate module (`mppt_top_wrapper`) was developed that converts each `integer` output into `std_logic_vector(31 downto 0)` through the `to_signed` function, and each `signed(15 downto 0)` input from `std_logic_vector` via a direct *cast*. This wrapper does not change the controller logic; it acts exclusively as a type bridge between the two language domains.

### B. Testbench architecture

Fig. 6 presents the hierarchical architecture of the developed UVM testbench. The environment follows the standard topology of the methodology: at the highest level, the *Test* component instantiates the *Environment*, which in turn contains an *Agent* and a *Scoreboard*. The *Agent* groups three components: the *Sequencer*, responsible for arbitrating the order of the transactions; the *Driver*, which converts each transaction into electrical stimuli applied to the DUT interface; and the *Monitor*, which passively observes the output signals and encapsulates the results into transaction objects. The connection between the *Monitor* and the *Scoreboard* is made through a TLM (*Transaction-Level Modeling*) analysis port, ensuring decoupling between observation and checking.

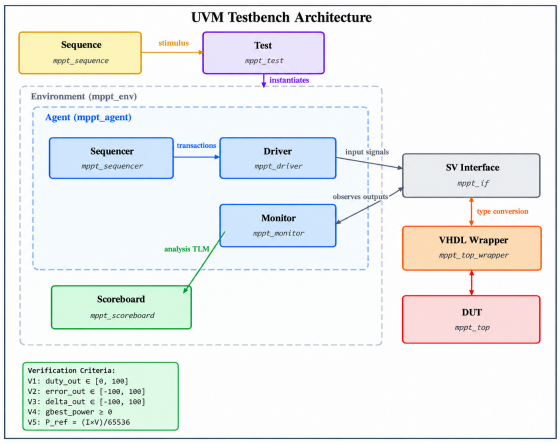



Figure 6. Hierarchical architecture of the UVM testbench for the hybrid MPPT controller, showing the transaction flow between the verification components, the connection to the VHDL DUT through the SystemVerilog interface, and the five reported Scoreboard criteria. V5 documents reference-power calculation; a corresponding DUT comparison is unconfirmed.

Each transaction is represented by an `mppt_seq_item` object that encapsulates the input fields (`current_in` and `voltage_in`, both 16-bit signed) and the output fields captured by the Monitor (`duty_out`, `gbest_power_out`, `error_out`, `delta_e_out`, among others).

### C. Verification criteria

The *Scoreboard* is described by five criteria for transactions received from the Monitor, as listed in Table IV. The first three criteria check that the controller outputs remain within the valid ranges defined by the VHDL implementation, where the saturation functions (`clamp`) enforce the limits. The

fourth criterion checks nonnegativity of the stored global-best power. For V5, the reference expression uses truncation toward zero, matching the DUT's integer division in Eq. 5. Here, $\mathrm{trunc}$ removes the fractional part toward zero, including for negative operands. The UVM Scoreboard is unavailable, so its division convention and sample-aligned DUT comparison remain unconfirmed. V5 is therefore a reference definition, not an established power-consistency assertion.

V1–V3 are also enforced by the VHDL subtypes of the corresponding signals (`duty_t`, 0 to 100; `err_t`, $-100$ to 100), and V4 by the clamping of negative power before the best-state update. A violation would therefore abort the simulation with a range error before reaching the Scoreboard, so these criteria confirm that no such abort occurred in the reported scenarios rather than an independent property of the arithmetic.

Table IV
REPORTED RANGE/SIGN CHECKS V1–V4 AND DUT-COMPATIBLE REFERENCE DEFINITION V5.

| ID | Criterion | Condition or calculation |
|---|---|---|
| V1 | Duty cycle range | $0 \leq \texttt{duty_out} \leq 100$ |
| V2 | Error signal range | $-100 \leq \texttt{error_out} \leq 100$ |
| V3 | Change-of-error range | $-100 \leq \texttt{delta_e_out} \leq 100$ |
| V4 | gbest nonnegativity | $\texttt{gbest_power_out} \geq 0$ |
| V5 | Reference power definition | $P_{ref} = \mathrm{trunc}(I_{in}V_{in}/65536)$ |

### D. Test scenarios

Twenty-eight test scenarios organized into seven categories were defined, as shown in Table V. The scenario selection exercises different regions of the input space, including normal conditions, numerical representation limits, and inputs relevant to saturation, overflow, and sign handling. No quantitative functional or code coverage is established by the available documentation, and output-range checks alone cannot exclude all intermediate arithmetic errors.

### E. Verification results

The UVM environment and its stimulus files are not part of the public repository, and the RTL revision and generic configuration used in the reported runs were not recorded with the results. The current RTL revision corrects the declared range of the raw $I \times V$ product in `mppt_measurement_unit`: with the previous upper limit of $2^{30} - 1$, the product of two inputs at $-32\,768$, which equals $2^{30}$, aborted the simulation and would wrap the sign in hardware. If Category 6 applied $-32\,768$ to both inputs simultaneously, the reported pass was obtained with an earlier revision. A directed test in the repository (`verification/tb_extremes.vhd`) now checks the signed extremes against $P_{ref}$, and an equivalence test against the original RTL (five generic configurations, 3,000 samples each) found no divergence. The UVM results below were not re-run on the current revision.

The simulation was run in ModelSim Intel FPGA Starter Edition 2021.1 with the UVM 1.2 library. Fig. 7 presents the duty cycle generated by the DUT for each of the 28 scenarios, grouped by category. The controller kept the duty cycle within the valid range $[0, 100]\%$ in all scenarios, including the positive boundary (Category 4) and negative boundary (Category 6) cases, where the input values approach the representation limits of the `signed(15:0)` type.

Fig. 8 compares the reference power calculated independently by the Scoreboard with the best global power (`gbest_power`) kept by PSO. In the scenarios with input products of negative sign in Categories 5 and 6, the reference power is negative, while `gbest_power` remains zero or positive, consistent with the nonnegativity criterion V4. The global-best state includes memory and filtering, so it is not an instantaneous DUT power result against which to assert equality with $P_{ref}$.

Fig. 9 presents the error (`error_out`) and change-of-error (`delta_e_out`) signals for all scenarios. Both remained within the interval $[-100, 100]$ in the reported transactions, supporting the range properties V2 and V3 under these stimuli without establishing the numerical accuracy of either signal.

Taken together, the reported results support the duty-cycle, error, and change-of-error range properties and global-best power nonnegativity for 28 scenarios across seven categories. This bounded functional evidence complements the synthesis results in Table III; it does not prove the complete PSO–Fuzzy algorithm correct. Section VII presents the complementary open-loop computational assessment.

## VI. STATIC TIMING ANALYSIS

After implementing the proposed architecture in VHDL, a Static Timing Analysis (STA) was performed using Intel Quartus Prime Lite Edition 20.1.1. The analysis aimed to characterize the timing behavior of the hybrid PSO–Fuzzy MPPT controller implementation on the Cyclone V 5CEBA4F23C7 device, used as the target device in the synthesis process.

The timing analysis considered the two clock signals used in the implementation, `clk_osc` and `clk_core`. The `clk_osc` signal was analyzed with a period of 20 ns, corresponding to the reference frequency of 50 MHz, while `clk_core` was analyzed with a period of 320 ns, corresponding to the reference frequency of 3.125 MHz.

The `clk_osc` corresponds to the 50 MHz oscillator of the DE0-CV development board targeted by the design and exclusively feeds the frequency divider of the `mppt_clock_reset_unit` module. The reported build contains no pin location assignments, so the Fitter placed all 44 I/O pins and the design was not mapped to or tested on the board; the I/O ports are excluded from the analysis by false-path constraints, since no external interface timing is specified. The `clk_core`, generated by this divider, clocks all of the controller logic: the state machine, the PSO swarm update, and the *fuzzy* engine. Its frequency is given by

$$f_{\mathrm{core}} = \frac{f_{\mathrm{osc}}}{N} = \frac{50\ \mathrm{MHz}}{16} = 3.125\ \mathrm{MHz}, \tag{6}$$

where $N$ is the division factor, defined by the generic parameter `CLK_DIV_G`.

Table V
TEST CATEGORIES AND SCENARIOS APPLIED TO THE MPPT CONTROLLER.

| Cat. | Name | Qty. | Objective |
|---|---|---|---|
| 1 | Null conditions | 3 | Behavior with zero inputs (total or partial absence of generation). |
| 2 | Normal operation | 5 | Typical current and voltage range, from very low to very high power. |
| 3 | I/V imbalance | 4 | Asymmetry between current and voltage, exercising the calculation of $\Delta P/\Delta V$. |
| 4 | Positive boundary | 3 | Values close to $+32\,767$ (maximum of `signed(15:0)`), testing overflow in the $I \times V$ multiplication. |
| 5 | Negative values | 6 | Negative current and/or voltage, producing negative or positive power (both negative). Checks that PSO does not track negative power. |
| 6 | Negative boundary | 3 | Values at $-32\,768$ (minimum representable), testing the worst-case overflow in signed arithmetic. |
| 7 | Abrupt transitions | 4 | Abrupt changes between extreme scenarios (maximum→zero, positive→negative), evaluating the dynamic response of the controller. |
| | **Total** | **28** | |

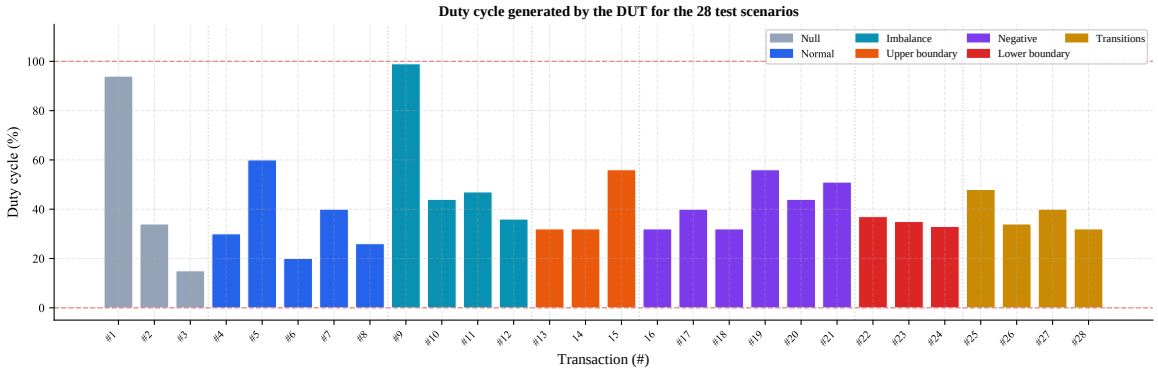



Figure 7. Duty cycle generated by the DUT for the 28 test scenarios, grouped by category. The red dashed lines indicate the valid limits $[0, 100]\%$. The dotted vertical lines separate the categories.

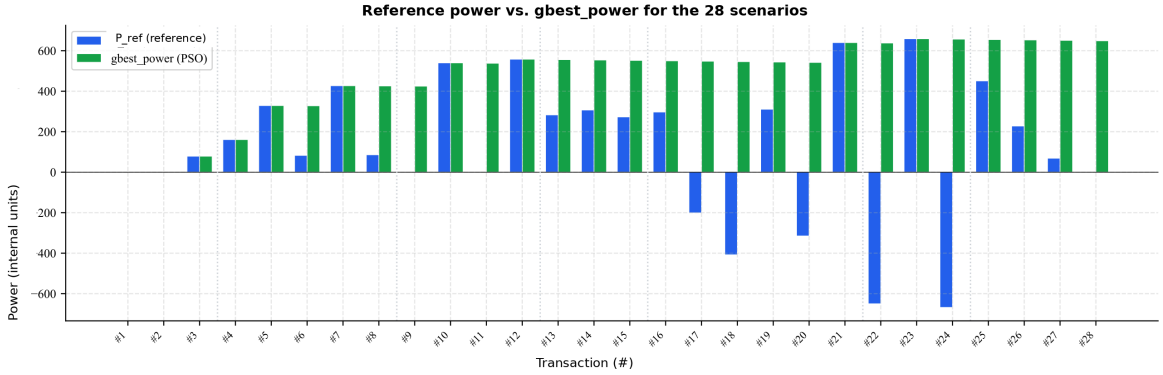



Figure 8. Comparison between the reference power ($P_{ref}$) and the best global PSO power (`gbest_power`) for the 28 scenarios. Nonnegative `gbest_power`, including when $P_{ref}$ is negative in Categories 5 and 6, supports V4 for the reported observations.

Table VI
STATIC TIMING ANALYSIS (STA) RESULTS.

| Corner | Clock | Period (ns) | Freq. (MHz) | Setup Slack (ns) | Fmax (MHz) | Hold Slack (ns) | Recovery Slack (ns) | Removal Slack (ns) | MPW Slack (ns) |
|---|---|---|---|---|---|---|---|---|---|
| Slow 1100mV 85C | `clk_osc` | 20 | 50.000 | 13.289 | 149.009 | 0.365 | – | – | 9.348 |
| Slow 1100mV 85C | `clk_core` | 320 | 3.125 | 154.681 | 6.049 | 0.381 | 313.832 | 2.150 | 159.339 |
| Slow 1100mV 0C | `clk_osc` | 20 | 50.000 | 13.349 | 150.353 | 0.342 | – | – | 9.235 |
| Slow 1100mV 0C | `clk_core` | 320 | 3.125 | 148.991 | 5.848 | 0.362 | 314.125 | 2.028 | 159.271 |
| Fast 1100mV 85C | `clk_osc` | 20 | 50.000 | 16.422 | 279.486 | 0.191 | – | – | 9.314 |
| Fast 1100mV 85C | `clk_core` | 320 | 3.125 | 240.554 | 12.587 | 0.178 | 315.971 | 1.222 | 159.657 |
| Fast 1100mV 0C | `clk_osc` | 20 | 50.000 | 16.540 | 289.017 | 0.169 | – | – | 9.289 |
| Fast 1100mV 0C | `clk_core` | 320 | 3.125 | 245.767 | 13.471 | 0.169 | 316.482 | 1.096 | 159.631 |

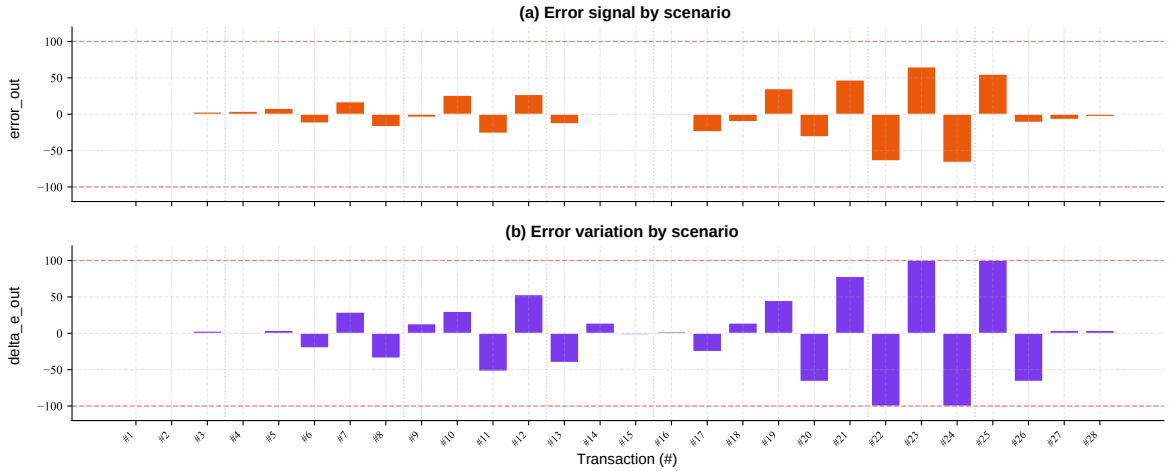



Figure 9. Error and change-of-error signals for the 28 test scenarios. (a) `error_out`. (b) `delta_e_out`. The red dashed lines indicate the limits $[-100, 100]$.

The reported timing results require a divided clock for the controller logic. In the worst-case `clk_core` corner (*Slow 1100mV 0C*), the reported $f_{\max} = 5.848$ MHz corresponds to a minimum period of approximately 171 ns, longer than the oscillator's 20 ns period. In all four corners, the worst `clk_core` setup path starts at the `prev_voltage` register and ends at the `pno_candidate` register of `hybrid_pso_fuzzy_mppt`. Within a single cycle, it traverses the voltage variation computed by `mppt_measurement_unit`, the *fuzzy* inference, the Fokker–Planck-inspired step, and the perturb-and-observe (P&O) direction logic of `mppt_fuzzy_ffp_unit`.

The factor $N = 16$ satisfies the divider structure and provides margin against the reported timing limit. First, $N$ must be even so that the counter toggles its output every $N/2$ oscillator cycles and produces a clock with a 50% duty cycle. Second, the resulting frequency remains below the worst-corner $f_{\max}$. With $N = 16$, the 320 ns period is about 1.87 times the required minimum period, consistent with the reported *setup slack* of 148.991 ns. Smaller dividers reduce this margin: $N = 8$ gives 6.25 MHz, above that $f_{\max}$, and $N = 10$ gives 5 MHz, only about 14.5% below it. This establishes timing feasibility under the analyzed constraints, not sufficient MPPT tracking speed. The controller update latency follows from the FSM schedule: one iteration over the ten particles takes $10\,\texttt{SETTLE_CYCLES} + 142$ cycles of `clk_core`, which a dedicated simulation confirmed, that is, 10,142 cycles or about 3.25 ms with `SETTLE_CYCLES` $= 1000$ at 3.125 MHz. Physical settling requires a converter and PV model or measurements.

To evaluate the timing behavior of the implementation under different conditions, the STA was run considering four timing corners: *Slow 1100mV 85C*, *Slow 1100mV 0C*, *Fast 1100mV 85C*, and *Fast 1100mV 0C*. In this way, the results allow the timing characteristics of the implementation to be compared across the evaluated conditions.

The analysis considered the timing margins provided by the tool, including *setup slack*, *hold slack*, *recovery slack*, *removal slack*, and *minimum pulse width slack* (MPW). The maximum operating frequency (*fmax*) obtained for each combination of clock signal and corner was also considered.

Table VI presents the results obtained in the static timing analysis for the different corners and clock signals considered.

For `clk_osc`, defined with a period of 20 ns and a reference frequency of 50 MHz, the lowest maximum frequency observed was 149.009 MHz, in the *Slow 1100mV 85C* corner. In the other corners, the maximum frequency remained above this value, reaching 289.017 MHz in the *Fast 1100mV 0C* corner. These values characterize the oscillator/divider domain, not the maximum operating frequency of the PSO–Fuzzy controller logic. The $f_{\max}$ column of Table VI is computed as $1/(T - \text{setup slack})$; for `clk_osc`, the worst setup path is the feedback from the divided-clock register (`clk_core_reg`) into the divider itself, so this value is more conservative than the Quartus *Fmax Summary*, which considers only paths launched and latched by the same clock (361.66 MHz for `clk_osc` in the *Slow 1100mV 85C* corner).

For `clk_core`, defined with a period of 320 ns and a reference frequency of 3.125 MHz, the lowest maximum frequency observed was 5.848 MHz, in the *Slow 1100mV 0C* corner. In the other scenarios, maximum frequency values between 6.049 MHz and 13.471 MHz were obtained.

The *setup slack* values presented in Table VI remain positive for all analyzed combinations. The *hold slack* values are also positive in all cases, with a minimum of 0.169 ns in the *Fast 1100mV 0C* corner. The *recovery*, *removal*, and *minimum pulse width* checks likewise show positive margins, the smallest value being the *removal slack* of 1.096 ns, also in the *Fast 1100mV 0C* corner.

Considering the maximum frequency results, the adopted frequencies of 50 MHz and 3.125 MHz correspond, respectively, to about 34% and 53% of the lowest *fmax* values obtained for `clk_osc` (149.009 MHz) and `clk_core` (5.848 MHz). The reported build therefore meets the analyzed

timing constraints in all evaluated corners, with positive margin in the reported checks. This does not establish external sensor or converter interface timing or physical settling time.

Together with the synthesis results presented earlier, the static timing analysis complements the characterization of the FPGA implementation.

## VII. Results and Discussion

### A. Description of the experiments

The experiments were carried out using the *Power, voltage, frequency and temperature dataset from Mesa Del Sol microgrid*, made available by Bashir et al. [29], the same dataset used by Pires and Oliveira [30].

The assessment used 15 monthly datasets, from May 2022 to July 2023, as preprocessed voltage and current input sequences. The dataset does not provide PV-side DC voltage or current. The voltage input is the AC voltage of the microgrid low-voltage main switchboard (`MG-LV-MSB_AC_Voltage`, about 480 V), and the current input is derived as $I = P_{PV}/V_{AC}$ from the PV inverter active power (`PVPCS_Active_Power`), normalized in each monthly file by its maximum. Consequently, the error $E = \Delta P/\Delta V$ computed from these inputs is not the slope of a PV array P–V curve, and power magnitudes are not comparable across months. The generated commands were not applied to a PV system or closed-loop electrical model, so subsequent inputs and input power were independent of the controller output. Daily aggregation characterizes generated commands and internal computational signals; it is not a measurement of MPPT tracking efficiency or energy extraction. The architectural default `SETTLE_CYCLES` $= 1000$ was replaced by 1 in the historical-data testbench, without modeling converter settling.

The hyperparameters of the hybrid controller were calibrated according to the procedure described in Section VII-B.

After hyperparameter selection, the hybrid PSO–Fuzzy controller was simulated over the historical input sequences. The following controller-signal metrics are reported daily:

- `duty_std_after_conv`: standard deviation of the generated `control_duty` from the threshold observation through the end of the day, rather than of the current-particle `duty` signal;
- `mean_abs_error`: daily mean absolute magnitude of the internal controller error signal, not error relative to a floating-point or high-precision reference.

For `duty_std_after_conv`, the interval starts at the first daily observation reaching 98% of the daily maximum of `power_now` and ends at the day's last observation. The whole day is used if the maximum is nonpositive or no valid threshold event is found. This input-based interval does not denote PSO convergence. The sample standard deviation returns zero for fewer than two valid observations; zero variability alone does not demonstrate command stabilization or control quality.

The VHDL description, experimental scripts, verification testbenches, and synthesis and timing reports are available in a public GitHub repository; the revision used in this paper is tagged `paper-v1`[1].

### B. Hyperparameter selection

The hyperparameters were chosen through an exploratory, limited search using the first five days of the April 2023 dataset as the calibration subset. The aim was to identify a configuration favored by a predefined open-loop command-stability objective, not a globally optimal MPPT configuration. The search included the main controller parameters and the adaptive-memory parameters of PSO; it did not evaluate extracted PV power or GMPP tracking quality.

The fuzzy rule matrix was kept fixed, using the rule base attributed to the reference fuzzy implementation by Pires and Oliveira [30]. This avoids varying the rule base during tuning, but does not independently isolate the effects of PSO and the Fokker–Planck-inspired stage. The initial PSO configuration uses coefficients $c_1 = 0.5$, $c_2 = 0.4$ and random-factor interval $\rho \in [0.53, 0.56]$ in the available source. These choices are motivated by [3]; exact correspondence with that reference is not established here. The configuration selected by the search, used in both the historical-data evaluation and the synthesized design, adopts $w = 0.30$, $c_1 = 0.30$, $c_2 = 0.50$ and $\rho \in [0.55, 0.65]$ (Table VII).

The procedure was divided into two stages. In the first, the parameters with the greatest direct impact on the controller dynamics were varied, such as the PSO weights, the dead zone, the search radius, the direction of action on the duty cycle, and the maximum step of the Fokker–Planck stage. In the second, the best results of the first stage were refined by varying more sensitive parameters, such as the velocity limits, the edge of the fuzzy membership functions, the range of $\rho$, and the parameters of the PSO adaptive memory. Stage 1 evaluated 20 randomly sampled configurations, and Stage 2 evaluated 8 random refinements around each of the three best Stage 1 configurations (24 trials), with a fixed random seed. Table VII summarizes the parameters considered and the selected configuration, which obtained the highest score ($S = -2.605$). Three Stage 2 trials, differing only in `RHO_MIN`, `VEL_MIN`, and the adaptive-memory and change-detection parameters, reached this score. Ties are broken deterministically, first preferring change detection disabled, which yields simpler hardware, and then the lowest trial index.

The score depends only on `control_duty` and on the error signals. In this open-loop setting, neither depends on the PSO state: the refined candidate is updated from its previous value and from the measurement-derived quantities through the fuzzy, Fokker–Planck-inspired, and P&O path, and the measurements do not depend on the applied command. The objective is therefore invariant to the PSO and adaptive-memory parameters. In all eight groups of trials that shared the same fuzzy and Fokker–Planck parameters (25 trials, each with a different PSO configuration), the score was identical,

[1] https://github.com/RamonPapes/FPGA_MPPT_FUZZY_PSO/tree/paper-v1

Table VII

HYPERPARAMETERS CONSIDERED IN THE EXPLORATORY TUNING PROCESS AND THE CONFIGURATION SELECTED BY THE SEARCH. DURING STAGE 1, THE STAGE 2 PARAMETERS WERE HELD AT RHO_MIN $= 53$, RHO_MAX $= 56$, VEL $= [-20, 20]$, FUZZY_EDGE $= 90$, AND CHANGE DETECTION DISABLED. PSO WEIGHTS AND $\rho$ ARE SCALED BY $1/100$ IN HARDWARE. †PSO AND ADAPTIVE-MEMORY PARAMETERS: THE OPEN-LOOP OBJECTIVE IS INVARIANT TO THEM, SO THEIR SELECTED VALUES ARE THOSE OF THE ADOPTED TRIAL AND ARE NOT IDENTIFIED BY THE SEARCH (SECTION VII-B).

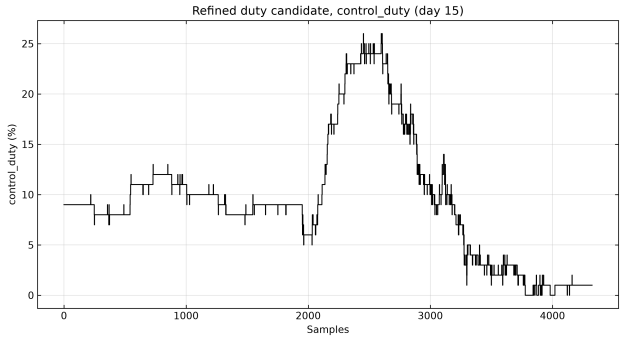
| Parameter | Status | Tested values | Selected | Rationale |
|---|---|---|---|---|
| $W_{\text{PSO}}$ | Stage 1 | 30, 40, 50, 60, 70 | 30† | Variation around the nominal value 0.50 to adjust the inertia of the particles. |
| $C_1$ | Stage 1 | 30, 40, 50, 60, 70 | 30† | Variation around the value 0.50 used as the reference for the cognitive term. |
| $C_2$ | Stage 1 | 20, 30, 40, 50, 60 | 50† | Variation around the value 0.40 used as the reference for the social term. |
| SEARCH_RADIUS | Stage 1 | 6, 8, 10, 12, 16, 20 | 6† | Controls the width of the local search space of the particles. |
| SEARCH_CENTER_MODE | Stage 1 | 0, 1, 2 | 1† | Tests the search center at pno_candidate, gbest, or the average of both. |
| DEADZONE | Stage 1 | 1, 2, 3, 4, 5 | 3 | Adjusts the sensitivity of the fuzzy controller to small error variations. |
| FOKKER_STEP_MAX | Stage 1 | 4, 6, 8, 10, 12 | 12 | Controls the maximum aggressiveness of the adjustment associated with the Fokker–Planck stage. |
| FUZZY_STEP | Stage 1 | 20, 30, 40 | 30 | Changes the width of the inner membership functions. |
| DUTY_DIRECTION | Stage 1 | -1, 1 | -1 | Tests the direction in which the controller acts on the duty cycle. |
| FUZZY_EDGE | Stage 2 | 70, 80, 90, 100 | 100 | Refines the opening of the outer membership functions. |
| VEL_MIN | Stage 2 | -30, -20, -10 | -30† | Adjusts the lower velocity limit of the particles. |
| VEL_MAX | Stage 2 | 10, 20, 30 | 10† | Adjusts the upper velocity limit of the particles. |
| RHO_MIN | Stage 2 | 50, 53, 55 | 55† | Refines the lower limit of $\rho$ in the reference-inspired PSO stage [3]. |
| RHO_MAX | Stage 2 | 56, 60, 65 | 65† | Refines the upper limit of $\rho$ in the reference-inspired PSO stage [3]. |
| MEMORY_HALF_LIFE | Stages 1–2 | 150, 300, 600, 900 (Stage 1); 200, 300, 450, 600, 900 (Stage 2) | 450† | Controls the exponential forgetting rate applied to the power value associated with pbest. |
| MAX_PBEST_AGE | Stages 1–2 | 250, 500, 1000 (Stage 1); 250, 500, 750, 1000 (Stage 2) | 500† | Limits for how many samples an old pbest can keep influencing the calculation of gbest. |
| ENABLE_CHANGE_DETECTION | Stage 2 | 0, 1 | 0† | Evaluates the optional use of abrupt-drop detection in operating conditions, without daily reset of the controller. |
| DROP_THRESHOLD_PERCENT | Stage 2 | 60, 70, 80 | 60† (inactive) | Defines the percentage threshold used by the optional abrupt-drop detection. |
| DROP_PATIENCE | Stage 2 | 20, 30, 60 | 20† (inactive) | Defines the number of consecutive samples required to confirm the optional abrupt-drop detection. |
| RESET_ON_DATE_CHANGE | Fixed | 0 | 0 | Kept disabled so that the controller is not automatically reset when the day changes. |
| FOKKER_STEP_MIN | Fixed | 1 | 1 | Kept fixed to avoid a null step in the duty cycle adjustment. |
| POWER_SCALE_DEN | Fixed | 65536 | 65536 | Sets the internal power scale for Q12.4 voltage and normalized Q1.15 current; it is not by itself a conversion to watts. |
| SETTLE_CYCLES | Fixed | 1 | 1 | Kept fixed to reduce the computational cost of the simulations; the synthesized design uses 1000. |
| FUZZY_RULES | Fixed | Fuzzy rule matrix | – | Kept identical to the reference fuzzy implementation [30]. |
| N_PARTICLES | Fixed | 10 | 10 | Kept fixed to avoid structural changes to the hardware during this comparison stage. |

and simulating the full April 2023 sequence with different PSO and memory settings left control_duty and the error signals bit-identical. The search thus selects only the dead zone, Fokker–Planck step, membership-function, and duty-direction parameters; the reported PSO and memory values († in Table VII) are those of the adopted trial and are not identified by the calibration.

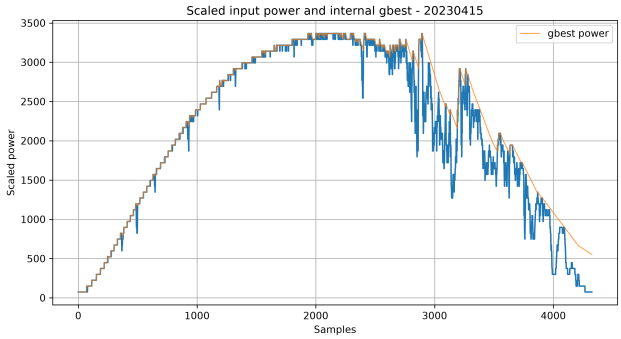
The optimizer ranks configurations using the composite score defined in Eq. (7). The terms and weights were chosen heuristically, without a formal optimization of the scoring function, to favor configurations with limited duty-cycle variability and saturation. The objective was to select a reasonable configuration for the open-loop computational evaluation, rather than to identify a globally optimal parameter set or demonstrate superior MPPT tracking performance.

$$\begin{aligned} S = -\big(&1.00\,\overline{\sigma_{D,\mathrm{f}}} + 0.30\,\overline{R_{D,\mathrm{f}}} + 0.20\,\overline{\sigma_D} \\ &+ 0.05\,\overline{R_D} + 0.50\,\overline{s_{D,\mathrm{s}}} \\ &+ 0.10\,\overline{m_{|E|}} + 0.10\,\overline{m_{|\Delta E|}} \\ &+ 20.0\,(1 - \overline{c_D}) \\ &+ 5.0\,\overline{v_{\mathrm{cons}}} + 5.0\,\overline{v_{\mathrm{mono}}}\big). \end{aligned} \tag{7}$$

Bars denote equally weighted averages over calibration days. For each day, $\sigma_D$ and $R_D$ are the standard deviation and range of control_duty; subscript f denotes the final window of up to $w = \min(200, \max(20, \lfloor n/10 \rfloor))$ samples, where $n$ is the daily sample count. The term $s_{D,\mathrm{s}}$ is the percentage of commands at the limits 0 or 100 over the




(a) Refined duty candidate (`control_duty`) on April 15.




(b) Input power and internal *gbest* evolution on April 15.

Figure 10. Open-loop computed signals on April 15. `control_duty` is a simulation-only observation signal, not an output of the synthesized top level; input power is independent of the generated duty cycle; *gbest* is an internal state.

Table VIII

DAILY $duty_{\text{STD_AFTER_CONV}}$: STANDARD DEVIATION OF GENERATED CONTROL DUTY AFTER THE OBSERVED-POWER THRESHOLD, OR OVER THE WHOLE DAY IF THE THRESHOLD IS UNDEFINED.

| Month/Year | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| May/2022 | 0.569 | 2.352 | 0.842 | 2.445 | 1.477 | 1.551 | 1.222 | 2.041 | 2.524 | 2.419 | 2.537 | 1.327 | 1.361 | 1.938 | 2.342 | 2.304 | 2.635 | 1.666 | 1.935 | 2.179 | 1.432 | 0.869 | 2.234 | 1.373 | 7.987 | 0.990 | 0.000 | 0.000 | 0.000 | 0.036 | 0.000 |
| Jun/2022 | 4.261 | 8.987 | 5.191 | 8.173 | 16.925 | 16.102 | 7.141 | 7.562 | 14.765 | 18.026 | 24.131 | 11.771 | 17.189 | 10.792 | 6.473 | 9.992 | 15.205 | 22.904 | 15.008 | 21.546 | 5.640 | 11.431 | 25.599 | 9.097 | 23.633 | 9.912 | 13.592 | 12.546 | 15.975 | 9.430 | |
| Jul/2022 | 9.253 | 10.126 | 3.721 | 18.390 | 8.574 | 16.093 | 24.098 | 27.275 | 9.581 | 9.912 | 10.189 | 15.391 | 10.952 | 23.151 | 16.299 | 10.833 | 9.656 | 12.266 | 3.981 | 8.864 | 9.264 | 1.719 | 19.304 | 7.668 | 9.090 | 8.140 | 10.313 | 10.123 | 23.780 | 18.943 | 9.946 |
| Aug/2022 | 1.999 | 6.940 | 11.138 | 12.740 | 10.035 | 1.289 | 1.270 | 7.314 | 10.304 | 6.243 | 10.255 | 25.474 | 11.644 | 4.540 | 25.183 | 18.492 | 10.134 | 9.219 | 16.599 | 11.378 | 15.989 | 9.122 | 6.942 | 3.711 | 9.098 | 10.000 | 16.213 | 26.658 | 9.328 | 13.487 | 8.694 |
| Sep/2022 | 11.300 | 15.525 | 10.261 | 17.024 | 7.665 | 11.895 | 8.870 | 18.009 | 7.719 | 8.417 | 7.287 | 5.721 | 5.404 | 9.335 | 17.312 | 11.372 | 8.041 | 6.453 | 10.526 | 11.970 | 13.570 | 3.912 | 6.831 | 10.997 | 5.658 | 13.659 | 0.000 | 19.388 | 14.705 | 18.404 | |
| Oct/2022 | 13.171 | 4.422 | 20.247 | 4.226 | 6.593 | 8.060 | 14.529 | 12.533 | 6.061 | 5.055 | 12.061 | 6.901 | 13.054 | 5.126 | 6.529 | 7.342 | 6.262 | 9.556 | 6.848 | 11.864 | 16.145 | 10.222 | 10.719 | 2.196 | 21.395 | 21.022 | 10.147 | 6.730 | 5.373 | 2.434 | 11.390 |
| Nov/2022 | 6.446 | 10.105 | 13.835 | 8.627 | 6.816 | 20.152 | 8.095 | 8.432 | 15.132 | 14.592 | 10.123 | 6.855 | 7.989 | 7.407 | 6.410 | 4.429 | 5.734 | 7.182 | 4.775 | 9.035 | 3.360 | 20.752 | 5.231 | 2.990 | 7.314 | 11.500 | 10.457 | 13.724 | 7.544 | 9.938 | |
| Dec/2022 | 8.727 | 5.959 | 9.463 | 10.542 | 0.000 | 4.110 | 1.905 | 0.646 | 3.235 | 5.665 | 8.105 | 7.716 | 10.112 | 13.888 | 3.302 | 17.884 | 8.917 | 0.000 | 3.584 | 7.313 | 9.948 | 8.014 | 17.839 | 0.000 | 5.808 | 16.844 | 11.638 | 5.999 | 12.839 | 9.960 | 3.988 |
| Jan/2023 | 12.747 | 10.127 | 12.534 | 2.697 | 9.447 | 4.901 | 39.805 | 1.077 | 6.812 | 5.589 | 27.503 | 24.295 | 2.531 | 16.881 | 7.193 | 20.330 | 14.452 | 15.122 | 22.559 | 7.211 | 4.922 | 15.827 | 12.473 | 4.712 | | | | | | | |
| Feb/2023 | 8.176 | 15.857 | 12.231 | 4.038 | 22.489 | 4.917 | 3.781 | 3.260 | 5.304 | 6.529 | 8.677 | 3.201 | 8.437 | 6.747 | 3.644 | 3.207 | 9.019 | 18.143 | 3.060 | 9.507 | 18.090 | 12.647 | 3.221 | 8.243 | 6.679 | 7.084 | 14.086 | 7.652 | | | |
| Mar/2023 | 5.950 | 3.539 | 4.540 | 0.000 | 27.699 | 4.222 | 4.576 | 6.253 | 8.943 | 9.689 | 14.056 | 14.991 | 5.206 | 2.101 | 32.305 | 8.999 | 6.137 | 13.688 | 15.851 | 6.563 | 14.810 | 6.875 | 15.293 | 18.409 | 14.411 | 11.230 | 10.358 | 8.232 | 7.359 | 5.990 | 7.929 |
| Apr/2023 | 16.773 | 1.223 | 1.361 | 3.554 | 1.378 | 1.494 | 5.188 | 2.939 | 1.269 | 4.246 | 1.533 | 0.859 | 1.679 | 1.438 | 8.326 | 0.680 | 1.389 | 2.724 | 1.030 | 1.207 | 5.885 | 2.071 | 0.399 | 1.153 | 1.480 | 0.693 | 1.542 | 1.723 | 10.879 | 2.503 | |
| May/2023 | 4.657 | 2.129 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | | 0.000 | 0.000 | 0.000 | 0.000 | | 0.000 | 0.000 | 0.000 | 0.000 | 0.030 | 0.797 | 0.083 | 0.256 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.632 | 0.139 | 0.000 | 0.065 | 0.723 |
| Jun/2023 | 11.411 | 2.594 | 3.113 | 13.065 | 10.763 | 4.742 | 11.488 | 7.003 | 2.132 | 11.421 | 0.377 | 10.652 | 0.000 | 6.089 | 7.978 | 7.862 | 0.000 | 0.000 | 7.500 | 5.187 | 9.123 | 11.022 | 5.077 | 5.556 | 5.883 | 6.754 | 5.106 | 22.667 | 15.098 | 6.798 | |
| Jul/2023 | 17.732 | 1.486 | 3.208 | 7.009 | 1.801 | 2.721 | 1.521 | 1.774 | 0.858 | 2.873 | 3.279 | 1.699 | 3.671 | 2.224 | 0.979 | 2.058 | 2.769 | 0.686 | 9.154 | 1.968 | 2.081 | 1.479 | 2.684 | 1.125 | 3.133 | 1.878 | 1.638 | 2.113 | 3.284 | 1.337 | 9.044 |

Table IX

DAILY $mean_{\text{ABS_ERROR}}$: MEAN ABSOLUTE INTERNAL CONTROLLER ERROR SIGNAL, NOT NUMERICAL ERROR AGAINST AN INDEPENDENT REFERENCE.

| Month/Year | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| May/2022 | 0.174 | 0.052 | 0.044 | 0.290 | 0.822 | 2.570 | 0.602 | 1.943 | 0.585 | 0.669 | 1.142 | 0.089 | 0.082 | 0.504 | 1.472 | 0.072 | 0.249 | 0.946 | 1.522 | 1.086 | 0.200 | 0.076 | 0.336 | 0.103 | 0.187 | 0.415 | 0.000 | 0.000 | 0.000 | 0.002 | 0.000 |
| Jun/2022 | 0.000 | 0.000 | 0.000 | 0.011 | 0.096 | 0.007 | 0.008 | 0.008 | 0.000 | 0.000 | 0.009 | 0.033 | 0.010 | 0.019 | 0.012 | 0.062 | 0.080 | 0.004 | 0.000 | 0.000 | 0.014 | 0.025 | 0.000 | 0.050 | 0.002 | 0.000 | 0.001 | 0.022 | 0.001 | 0.003 | |
| Jul/2022 | 0.038 | 0.052 | 0.029 | 0.006 | 0.002 | 0.003 | 0.000 | 0.009 | 0.000 | 0.048 | 0.032 | 0.040 | 0.013 | 0.008 | 0.004 | 0.126 | 0.093 | 0.093 | 0.010 | 0.013 | 0.006 | 0.003 | 0.007 | 0.103 | 0.052 | 0.018 | 0.001 | 0.058 | 0.093 | 0.000 | 0.047 |
| Aug/2022 | 0.005 | 0.019 | 0.000 | 0.002 | 0.005 | 0.011 | 0.075 | 0.168 | 0.077 | 0.024 | 0.010 | 0.033 | 0.032 | 0.061 | 0.024 | 0.006 | 0.007 | 0.012 | 0.006 | 0.000 | 0.003 | 0.036 | 0.052 | 0.155 | 0.115 | 0.081 | 0.074 | 0.010 | 0.019 | 0.000 | 0.000 |
| Sep/2022 | 0.004 | 0.004 | 0.013 | 0.009 | 0.000 | 0.000 | 0.059 | 0.022 | 0.078 | 0.003 | 0.033 | 0.010 | 0.023 | 0.052 | 0.028 | 0.053 | 0.086 | 0.051 | 0.005 | 0.003 | 0.019 | 0.037 | 0.000 | 0.000 | 0.000 | 0.000 | 0.308 | 0.000 | 0.000 | 0.000 | |
| Oct/2022 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.001 | 0.000 | 0.010 | 0.006 | 0.057 | 0.000 | 0.000 | 0.006 | 0.066 | 0.015 | 0.131 | 0.000 | 0.008 | 0.001 | 0.000 | 0.000 | 0.000 | 0.002 | 0.012 | 0.062 | 0.047 | 0.001 | 0.069 | 0.045 | 0.164 | 0.000 |
| Nov/2022 | 0.000 | 0.000 | 0.130 | 0.073 | 0.085 | 0.000 | 0.000 | 0.001 | 0.000 | 0.002 | 0.000 | 0.000 | 0.011 | 0.012 | 0.020 | 0.169 | 0.000 | 0.001 | 0.007 | 0.000 | 0.005 | 0.002 | 0.000 | 0.002 | 0.071 | 0.000 | 0.003 | 0.000 | 0.003 | 0.016 | |
| Dec/2022 | 0.002 | 0.000 | 0.016 | 0.032 | 0.000 | 0.000 | 0.000 | 0.000 | 0.003 | 0.002 | 0.028 | 0.010 | 0.012 | 0.000 | 0.000 | 0.001 | 0.000 | 0.003 | 0.001 | 0.028 | 0.044 | 0.004 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.010 | 0.000 | 0.000 |
| Jan/2023 | 0.003 | 0.003 | 0.000 | 0.000 | 0.000 | 0.000 | 0.005 | 0.011 | 0.005 | 0.000 | 0.000 | 0.008 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.002 | 0.000 | 0.000 | 0.002 | 0.019 | 0.001 | | | | | | | |
| Feb/2023 | 0.124 | 0.000 | 0.017 | 0.075 | 0.000 | 0.000 | 0.060 | 0.013 | 0.010 | 0.052 | 0.013 | 0.011 | 0.003 | 0.004 | 0.007 | 0.017 | 0.001 | 0.009 | 0.048 | 0.000 | 0.000 | 0.003 | 0.011 | 0.000 | 0.008 | 0.000 | 0.077 | 0.046 | | | |
| Mar/2023 | 0.022 | 0.125 | 0.023 | 0.002 | 0.000 | 0.015 | 0.085 | 0.005 | 0.000 | 0.000 | 0.004 | 0.000 | 0.042 | 0.035 | 0.132 | 0.000 | 0.018 | 0.000 | 0.000 | 0.002 | 0.001 | 0.031 | 0.004 | 0.000 | 0.000 | 0.109 | 0.132 | 0.007 | 0.000 | 0.012 | 0.004 |
| Apr/2023 | 0.232 | 0.674 | 1.036 | 0.434 | 0.842 | 1.556 | 0.206 | 0.079 | 0.305 | 0.138 | 0.320 | 0.649 | 0.972 | 0.643 | 0.322 | 0.246 | 0.077 | 0.426 | 0.281 | 0.677 | 0.185 | 1.321 | 0.204 | 0.936 | 0.646 | 0.395 | 2.453 | 1.304 | 0.092 | 0.386 | |
| May/2023 | 0.656 | 0.110 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | | 0.000 | 0.000 | 0.000 | 0.000 | | 0.000 | 0.000 | 0.000 | 0.000 | 0.002 | 0.019 | 0.011 | 0.008 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.026 | 0.056 | 0.000 | 0.009 | 0.034 |
| Jun/2023 | 0.439 | 0.556 | 0.843 | 0.840 | 0.526 | 0.590 | 0.435 | 0.471 | 0.085 | 0.714 | 0.047 | 0.755 | 0.000 | 0.681 | 0.514 | 0.581 | 0.000 | 0.000 | 0.352 | 0.787 | 0.526 | 0.792 | 0.795 | 0.615 | 0.817 | 0.640 | 0.559 | 0.631 | 0.734 | 0.922 | |
| Jul/2023 | 0.232 | 0.280 | 0.111 | 0.133 | 0.286 | 0.241 | 0.061 | 0.492 | 0.198 | 0.174 | 0.203 | 0.185 | 0.608 | 0.165 | 0.218 | 0.203 | 0.111 | 0.059 | 0.117 | 0.124 | 0.266 | 0.155 | 0.429 | 0.037 | 0.069 | 0.124 | 0.643 | 0.687 | 0.845 | 0.246 | 0.630 |

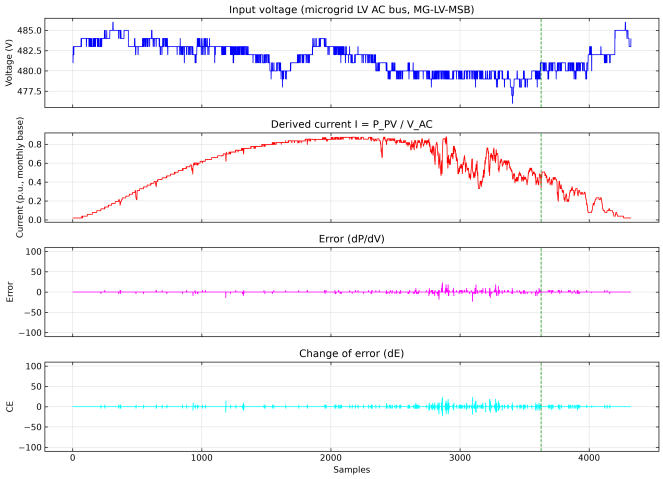



Figure 11. Preprocessed inputs (microgrid AC bus voltage and current derived from PV active power), internal error, and change of error on April 15, the same day evaluated by Pires and Oliveira [30].

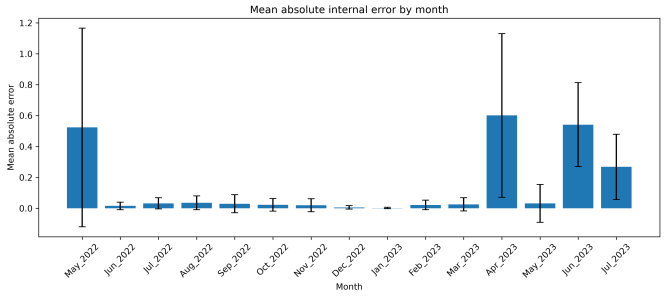



Figure 12. Monthly summary of the mean absolute internal controller error signal; this is not numerical error against an independent reference.

detected stable tail, or the final window if stabilization is not detected. The quantities $m_{|E|}$ and $m_{|\Delta E|}$ are daily mean absolute internal error and change of error, and $c_D$ is one when the duty-stabilization criterion is met and zero otherwise.

The fields $v_{\text{cons}}$ and $v_{\text{mono}}$ are defined as daily global-best consistency and monotonicity violation counts. For adaptive-

memory logs, the post-processing script assigns both zero without checking these properties. Thus, $\overline{v_{\mathrm{cons}}} = \overline{v_{\mathrm{mono}}} = 0$: their penalties were inactive in the reported adaptive-memory experiments and did not influence parameter selection. These zeros are not evidence of verification. The remaining heuristic terms rank open-loop command and internal-signal behavior.

### C. Open-loop behavior and descriptive metrics

Figs. 11 and 10 illustrate the behavior of the hybrid PSO–Fuzzy controller on April 15, the same day analyzed by Pires and Oliveira [30].

Fig. 11 presents the preprocessed AC bus voltage and derived current inputs together with the computed error and change of error. The fuzzy inputs vary with the historical measurements, particularly during intervals with greater current variation. The available files do not provide a matched baseline trace from Pires and Oliveira [30]; using the same calendar day alone does not establish reduced error peaks or lower command variability relative to that controller.

Fig. 10 presents the computed signals for the same day. Fig. 10a shows the refined duty candidate `control_duty`, while Fig. 10b shows the input-power signal and internal $gbest$ power. The former is a characteristic of the historical input sequence; the latter is an adaptive-memory state updated during execution. Their evolution does not associate any candidate duty cycle with physically obtained power.

In this open-loop evaluation, the `control_duty` trace reflects only the fuzzy, Fokker–Planck-inspired, and P&O refinement path. The PSO state, that is, particle positions and velocities, *pbest*, and *gbest*, does not feed back into the refined candidate, because the inputs do not depend on the applied command (Section VII-B). The PSO stage affects only the particle commands on `duty_out` and the internal *gbest* trace of Fig. 10b. Consequently, the duty-cycle and error results characterize the refinement path rather than the hybrid PSO–Fuzzy interaction, which requires closed-loop evaluation. This refinement differs structurally from direct fuzzy-output adjustment by $\Delta D$, as described for the reference implementation [30], but the present plots do not establish comparative tracking performance. Duty variability and any command-stabilization criterion describe generated signals; low variability alone is not proof of control quality.

Next, the analysis was extended to all available days in all months of the dataset by collecting the metrics defined above. The daily metrics are presented in Tables VIII and IX.

For internal computational behavior, Fig. 12 shows higher mean absolute controller error and greater dispersion in months such as May/2022, Apr/2023, Jul/2023, and Jun/2023. Other months have values close to zero. These magnitudes describe the implemented error signal, which is affected by fixed-point scaling, integer division, saturation, and explicit zeroing for small voltage changes. They do not measure fixed-point error against an independent reference or establish numerical accuracy or stability. The causes of the month-to-month differences remain for investigation.

The historical-data results characterize generated-command variability and internal signal magnitudes. Hardware feasibility is supported separately by the synthesis and STA results, and the UVM results support the specified range and sign properties under the reported scenarios. Together these provide implementation evidence and an open-loop computational characterization, while physical GMPP tracking and photovoltaic power extraction remain untested.

## VIII. Concluding Remarks

This work presented the design, fixed-point VHDL implementation, and characterization of a hybrid MPPT controller integrating particle swarm optimization (PSO), fuzzy logic, and Fokker–Planck-inspired adaptive stepping [3]. The modular architecture separates datapath and control logic. The reported synthesis for the Cyclone V 5CEBA4F23C7 device used 3,535 ALMs (19%) and 9 of 66 DSP blocks (14%), without block memory or PLLs, supporting hardware feasibility and resource characterization. Numerical scaling, adaptive memory with exponential forgetting, and implementation scheduling were documented as distinct adaptations and decisions rather than asserted as original algorithmic innovations.

The reported UVM assessment covered 28 scenarios and five specified criteria. Its results support the duty-cycle, error, and change-of-error ranges and nonnegative global-best power for the tested observations; V5 documents reference-power calculation, with a corresponding DUT comparison unconfirmed. These checks do not establish complete algorithmic correctness, and they were not re-run on the current RTL revision, which corrects the range of the raw $I \times V$ product for inputs at $-32\,768$. STA across four operating corners reports positive timing margins at the configured `clk_core` frequency of 3.125 MHz, with worst-case $f_{\max} = 5.848$ MHz, and one control iteration takes about 3.25 ms with `SETTLE_CYCLES` $= 1000$. Frequencies above 149 MHz belong to the oscillator/divider domain and do not characterize the main controller logic or establish MPPT tracking speed. The synthesis and timing tables were obtained from a single Quartus build of the current RTL revision, with the hyperparameter configuration also used in the historical-data evaluation.

The complementary assessment used 15 months of historical inputs to characterize generated duty-cycle variability and internal error signals. Low mean absolute internal error is not an independent measure of fixed-point numerical accuracy, and low command variability does not establish successful tracking. Exploratory tuning with the repository's `optimize_hyperparameters.py` script favored a command-stability objective and does not establish optimal MPPT parameters. In open loop, the PSO state does not feed back into the refined command, so the duty-cycle and error results, and the tuning objective itself, characterize only the fuzzy, Fokker–Planck-inspired, and P&O refinement path; the PSO and adaptive-memory parameters are not identified by the calibration.

The historical-data evaluation was exclusively open loop, without a PV array and DC–DC converter model or physical plant responding to the generated duty cycle. Consequently, it does not establish physical GMPP convergence, closed-loop tracking efficiency, or improvements in photovoltaic power extraction. The contributions are the implemented fixed-point architecture, reported synthesis and timing characterization, evidence for specified functional properties, and descriptive open-loop computational results. Closed-loop evaluation and controlled comparisons remain future work.

### A. Next steps

- **Extended UVM functional verification:** expand the verification environment with an independent reference model to validate PSO particle updates, fuzzy inference, adaptive *pbest*/*gbest* tracking, and FSM transitions. Introduce functional coverage metrics and additional test scenarios to assess numerical correctness beyond the currently verified output ranges.
- **Closed-loop validation and comparative evaluation:** integrate the controller with a photovoltaic array and DC–DC converter model or physical prototype, enabling duty-cycle feedback and evaluation of GMPP tracking efficiency, convergence time, and power oscillations under uniform irradiance and partial shading. Compare its performance with the reference implementation [30] and the algorithm proposed by Patnaik et al. [3] under equivalent operating conditions.
- **Hardware characterization and optimization:** extend FPGA evaluation by measuring controller update latency on the board with real sensor and converter interfaces, analyzing fixed-point numerical accuracy against a high-precision reference, and investigating resource utilization and timing trade-offs. Further explore hyperparameter sensitivity to assess its impact on computational behavior and hardware requirements.

These steps would extend the implementation and its public-data assessment toward closed-loop operation, controlled comparisons with reference MPPT controllers, and further hardware characterization. Publishing the UVM environment and re-running it on the current RTL revision would also strengthen the reproducibility of the verification results [29].

## REFERENCES


[1] T. M. d. Lima and J. A. C. B. Oliveira, "FPGA-based fuzzy logic controllers applied to the MPPT of PV panels—a systematic review," *IEEE Transactions on Fuzzy Systems*, vol. 32, no. 8, pp. 4260–4269, Aug. 2024.

[2] M. J. N. Cerqueira and J. A. Castelo Branco, "A systematic literature review of fpga-based mppt techniques for photovoltaic systems under partial shading," Manuscript, Universidade SENAI CIMATEC, 2026, paper in preparation.

[3] B. Patnaik, S. C. Swain, A. Ballaji, and R. Dash, "Design and analysis of mppt using advanced pso based on fuzzy fokker planck solution under partial shading condition," in *2022 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)*. IEEE, 2022, pp. 1–6.

[4] F. Belhachat and C. Larbes, "A review of global maximum power point tracking techniques of photovoltaic system under partial shading conditions," *Renewable and Sustainable Energy Reviews*, vol. 92, pp. 513–553, 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1364032118303149

[5] Mohd Amran Mohd Radzi, "Facing energy trilemma with solar power generation as future main player in infrastructure development," *IOP Conference Series: Materials Science and Engineering*, vol. 955, no. 1, p. 012068, 2020.

[6] S. Aly and I. Vrana, "Toward efficient modeling of fuzzy expert systems: A survey," *Agricultural Economics*, vol. 52, no. 10, pp. 456–460, 2006.

[7] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proceedings of ICNN'95 - International Conference on Neural Networks*, vol. 4, 1995, pp. 1942–1948.

[8] D. Andrade, C. Silva, and P. Mendes, "A critical analysis of PSO and its variations applied to MPPT for PV systems under partial shading condition," *Renewable Energy*, vol. 164, pp. 734–746, Feb. 2021.

[9] N. Hanafiah, M. R. Othman, and M. Arifin, "A hybrid MPPT for quasi-Z-source inverters in PV applications under partial shading condition," *International Journal of Power Electronics and Drive Systems*, vol. 7, no. 3, pp. 932–940, Sep. 2016.

[10] A. Sai, M. Kumar, and A. Reddy, "An improved MPPT technique under partial shading condition using simple P&O algorithm," *International Journal of Advanced Science and Technology*, vol. 29, no. 6, pp. 8571–8582, 2020.

[11] O. Oufettoul, A. Djerioui, and S. Drid, "Comprehensive analysis of MPPT control approaches under partial shading condition," in *IEEE International Conference on Renewable Energy Research and Applications (ICRERA)*. Piscataway: IEEE, Nov. 2022, pp. 165–170.

[12] M. A. Dirmawan, A. Hidayat, and S. R. Sulistyo, "The comparison performance of MPPT perturb and observe, fuzzy logic controller, and flower pollination algorithm in normal and partial shading condition," in *IEEE Industrial Electronics Seminar (IES)*. Piscataway: IEEE, 2020, pp. 102–108.

[13] Q. Duan *et al.*, "An improved variable step P&O and global scanning MPPT method for PV systems under partial shading condition," *Energies*, vol. 8, no. 6, pp. 5338–5354, 2015.

[14] R. Alik and N. M. Nor, "An improved perturb and observe checking algorithm MPPT for photovoltaic system under partial shading condition," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 18, no. 2, pp. 938–945, May 2020.

[15] Y. M. Safarudin *et al.*, "Combining simplified firefly and modified P&O algorithm for MPPT of PV system under partial shading condition," in *IEEE Conference on Sustainable Energy*. Piscataway: IEEE, 2015, pp. 180–186.

[16] M. Singh, R. Pandey, and P. Pandey, "Modified ABC–WOA algorithm under partial shading condition for MPPT in PV system," *Renewable Energy Focus*, vol. 46, pp. 109–120, 2023.

[17] R. Mahendran and S. Vijayakumar, "Modified grey wolf optimization algorithm for PV-fed water pumping system driven by switched reluctance motor under partial shading condition," *International Journal of Renewable Energy Research*, vol. 11, no. 4, pp. 1702–1712, 2021.

[18] Y. Wang and X. Chang, "Maximum power point tracking of photovoltaic array under partial shaded conditions based on improved grey wolf algorithm," *Renewable Energy*, vol. 210, pp. 1011–1021, Apr. 2023.

[19] R. Raj and A. Shukla, "Enhanced grey wolf optimization for GMPP tracking of PV systems under partial shading condition," *Solar Energy*, vol. 231, pp. 38–49, Oct. 2022.

[20] N. F. Baso *et al.*, "An improvement GWO–P&O algorithm based MPPT for PV system under partial shading condition," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 23, no. 2, pp. 842–850, Feb. 2024.

[21] M. K. Senapati and G. Tripathy, "PV system with binary grey wolf optimization-based MPPT for partial shading condition," in *IEEE International Symposium on Sensors, Systems and Control (iSSSC)*. Piscataway: IEEE, 2022, pp. 1–6.

[22] IEEE, *IEEE Standard VHDL Language Reference Manual*, Institute of Electrical and Electronics Engineers, 2009, revision of IEEE Std 1076-2002.

[23] K. Ishaque, Z. Salam, M. Amjad, and S. Mekhilef, "An improved particle swarm optimization (pso)–based mppt for pv with reduced steady-state oscillation," *IEEE Transactions on Power Electronics*, vol. 27, no. 8, pp. 3627–3638, 2012.

[24] T. Esram and P. L. Chapman, "Comparison of photovoltaic array maximum power point tracking techniques," *IEEE Transactions on Energy Conversion*, vol. 22, no. 2, pp. 439–449, 2007.

[25] B. Subudhi and R. Pradhan, "A comparative study on maximum power point tracking techniques for photovoltaic power systems," *IEEE Transactions on Sustainable Energy*, vol. 4, no. 1, pp. 89–98, 2013.

[26] E. H. Mamdani and S. Assilian, "An experiment in linguistic synthesis with a fuzzy logic controller," *International Journal of Man-Machine Studies*, vol. 7, no. 1, pp. 1–13, 1975.

[27] P. Alfke, *Efficient Shift Registers, LFSR Counters, and Long Pseudo-Random Sequence Generators*, Xilinx, Jul. 1996, revision 1.1. [Online]. Available: https://docs.amd.com/v/u/en-US/xapp052

[28] Accellera Systems Initiative, *Universal Verification Methodology (UVM) 1.2 Reference Manual*, 2014. [Online]. Available: https://www.accellera.org/downloads/standards/uvm

[29] A. Bashir, C. Leap, A. Blumenthal, T. Estrada, A. Bidram, M. Martinez-Ramon, and M. Abdullah, "Power, voltage, frequency and temperature dataset from Mesa Del Sol microgrid," Dataset, 2023. [Online]. Available: https://doi.org/10.5061/dryad.fqz612jzb

[30] P. B. Pires and J. A. C. B. Oliveira, "A parameterizable fuzzy mppt controller implemented in fpga-based photovoltaic systems," Undergraduate thesis, Universidade SENAI CIMATEC, 2025.

[31] World Meteorological Organization (WMO), "Previsão de condições climáticas extremas pode apoiar transição energética," UN News, New York, Mar. 2025, in Portuguese. [Online]. Available: https://news.un.org/pt/story/2025/03/1845786

[32] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Keele University and Durham University, EBSE Technical Report EBSE-2007-01, 2007.

[33] International Energy Agency, "Tracking SDG7: The energy progress report 2023," IEA, Paris, Tech. Rep., 2023.

[34] B. Hajji, A. Mellit, and L. Bouselham, "FPGA applications in renewable energy systems: Photovoltaic, wind-turbine and hybrid systems," in *A Practical Guide for Simulation and FPGA Implementation of Digital Design*. Singapore: Springer Nature Singapore, 2022, pp. 179–212.

[35] M. D. S. Kumar, P. S. Kumar, and R. Prasad, "FPGA-based real-time implementation of maximum power point tracking for photovoltaic system," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 5, pp. 3393–3402, May 2021.

[36] S. K. Kollimalla, M. P. Rajesh, and N. R. Babu, "A review on FPGA-based control techniques for renewable energy applications," *IEEE Access*, vol. 9, pp. 14 483–14 497, Feb. 2021.

[37] R. Verma, P. Singh, and P. Kumar, "A comparative study of MPPT techniques for photovoltaic system," *International Journal of Engineering Research & Technology*, vol. 10, no. 8, pp. 1–5, Aug. 2021.

[38] N. Kumar, B. Singh, and B. K. Panigrahi, "Partial shading mitigation in solar photovoltaic systems: A review," *IEEE Transactions on Sustainable Energy*, vol. 11, no. 4, pp. 2805–2816, 2020.

[39] H. Cangi, S. Adak, and A. S. Yilmaz, "Effect of temperature on the I–V and P–V curves of the photovoltaic cell," *Journal of the Institute of Science and Technology*, vol. 11, no. 4, pp. 2682–2694, 2021.

[40] L. Bhukya, N. R. Kedika, and S. R. Salkuti, "Enhanced maximum power point techniques for solar photovoltaic system under uniform insolation and partial shading conditions: A review," *Algorithms*, vol. 15, no. 10, p. 365, Oct. 2022.

[41] A. M. Eltamaly, M. S. Al-Saud, and A. M. Al-Saud, "An improved cuckoo search algorithm for maximum power point tracking of photovoltaic systems under partial shading conditions," *Energies*, vol. 14, no. 4, p. 953, 2021.

[42] S. Mirjalili, S. M. Mirjalili, and A. Lewis, "Grey wolf optimizer," *Advances in Engineering Software*, vol. 69, pp. 46–61, 2014.

[43] S. Brown and Z. Vranesic, *Fundamentals of Digital Logic with VHDL Design*. New York: McGraw-Hill, 2013.

[44] C. Maxfield, *The Design Warrior's Guide to FPGAs*. Burlington: Elsevier/Newnes, 2004.

[45] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*. Hoboken: Wiley, 2003.

[46] M. Kermadi, M. Hattab, S. Berkani, and S. Bouchakour, "Recent developments of MPPT techniques for PV systems under partial shading conditions: A critical review and performance evaluation," *IET Renewable Power Generation*, vol. 15, no. 12, pp. 2594–2610, 2021.

[47] J. Ahmed and Z. Salam, "A critical evaluation of maximum power point tracking methods for partial shading in photovoltaic systems," *Renewable and Sustainable Energy Reviews*, vol. 47, pp. 933–953, Jul. 2015.

[48] T. Esram and P. L. Chapman, "Comparison of photovoltaic array maximum power point tracking techniques," *IEEE Transactions on Energy Conversion*, vol. 22, no. 2, pp. 439–449, Jun. 2007.